

Estimation Theory Overview

- Properties
- Bias, Variance, and Mean Square Error
- Cramér-Rao lower bound
- Maximum likelihood
- Consistency
- Confidence intervals
- Properties of the mean estimator
- Properties of the variance estimator
- Examples

Introduction

- Up until now we have defined and discussed properties of random variables and processes
- In each case we started with some known property (e.g. autocorrelation) and derived other related properties (e.g. PSD)
- In practical problems we rarely know these properties apriori
- In stead, we must *estimate* what we wish to know from finite sets of measurements

Terminology

- Suppose we have N observations $\{x(n)\}_{n=0}^{N-1}$ collected from a WSS stochastic process
- This is *one realization* of the random process $\{x(n, \zeta)\}_{n=0}^{N-1}$
- Ideally we would like to know the joint PDF

$$f(x_1, x_2, \dots, x_n; \theta_1, \theta_2, \dots, \theta_p)$$

- Here θ are unknown parameters of the joint PDF
- In probability theory, we think about the likeliness of $\{x(n)\}_{n=0}^{N-1}$ given the PDF and θ
- In inference, we are given $\{x(n)\}_{n=0}^{N-1}$ and are interested in the likeliness of θ
- Called the **sampling distribution**
- We will use θ to denote a scalar parameter (or θ for a vector of parameters) we wish to estimate

Estimators as Random Variables

- Our **estimator** is a function of the measurements $\hat{\theta} \left[\{x(n)\}_{n=0}^{N-1} \right]$
- It is therefore a *random variable*
- It will be different for every different set of observations
- It is called an estimate or, if θ is a scalar, a **point estimate**
- Of course we want $\hat{\theta}$ to be as close to the true θ as possible

Natural Estimators

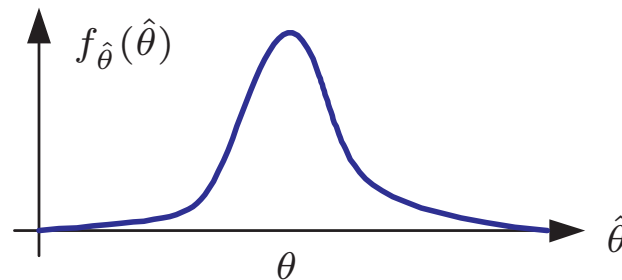
$$\hat{\mu}_x = \hat{\theta} \left[\{x(n)\}_0^{N-1} \right] = \frac{1}{N} \sum_{n=0}^{N-1} x(n)$$

- This is the obvious or “natural” estimator of the process mean
- Sometimes called the **average** or **sample mean**
- It will also turn out to be the “best” estimator
- I will define “best” shortly

$$\hat{\sigma}_x^2 = \hat{\theta} \left[\{x(n)\}_0^{N-1} \right] = \frac{1}{N} \sum_{n=0}^{N-1} [x(n) - \hat{\mu}_x]^2$$

- This is the obvious or “natural” estimator of the process variance
- Not the “best”

Good Estimators



- What is a “good” estimator?
 - Distribution of $\hat{\theta}$ should be centered at the true value
 - Want the distribution to be as narrow as possible
- Lower-order moments enable coarse measurements of “goodness”

Bias

Bias of an estimator $\hat{\theta}$ of a parameter θ is defined as

$$B(\hat{\theta}) \triangleq E[\hat{\theta}] - \theta$$

Normalized Bias of an estimator $\hat{\theta}$ of a non-negative parameter θ is defined as

$$\varepsilon_b \triangleq \frac{B(\hat{\theta})}{\theta}$$

- **Unbiased:** an estimator is said to be unbiased if $B(\hat{\theta}) = 0$
- This implies the PDF of the estimator is centered at the true value θ
- The sample mean is unbiased
- The estimator of variance on the earlier slide is biased
- Unbiased estimators are generally good, but they are not always best (more later)

Variance

Variance of an estimator $\hat{\theta}$ of a parameter θ is defined as

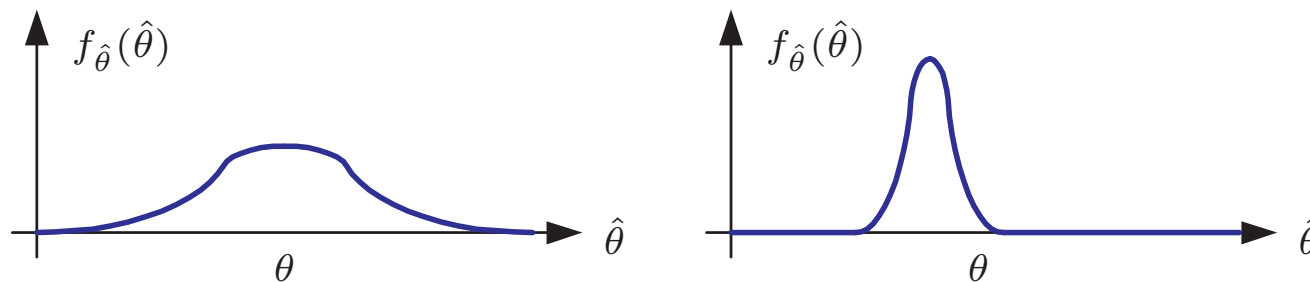
$$\text{var}(\hat{\theta}) = \sigma_{\hat{\theta}}^2 \triangleq \text{E} \left[\left| \hat{\theta} - \text{E}[\hat{\theta}] \right|^2 \right]$$

Normalized Standard deviation of an estimator $\hat{\theta}$ of a non-negative parameter θ is defined as

$$\varepsilon_r \triangleq \frac{\sigma_{\hat{\theta}}}{\theta}$$

- More commonly called the **coefficient of variation**
- A measure of the *spread* of $\hat{\theta}$ about *its mean*
- Would like the variance to be as small as possible

Bias-Variance Tradeoff



- In many cases minimizing variance conflicts with minimizing bias
- Note that $\hat{\theta} \triangleq 0$ has zero variance, but is generally biased
- In these cases we must trade variance for bias (or vice versa)

Mean Square Error

Mean Square Error of an estimator $\hat{\theta}$ of a parameter θ is defined as

$$\text{MSE}(\theta) \triangleq \text{E} \left[|\hat{\theta} - \theta|^2 \right] = \sigma_{\hat{\theta}}^2 + |B(\hat{\theta})|^2$$

Normalized MSE of an estimator $\hat{\theta}$ of a parameter θ may be defined as

$$\varepsilon \triangleq \frac{\text{MSE}(\hat{\theta})}{\sigma_{\theta}^2}$$

- This is different from the text
- The decomposition of MSE into variance plus bias squared is very similar to the DC and AC decomposition of signal power
- We will use MSE as a global measure of estimator performance
- Note that two different estimators may have the same MSE, but different bias and variance

- This criterion is convenient for building estimators
- Creating a problem we can solve

Cramér-Rao Lower Bound

$$\text{var}(\hat{\theta}) \geq -\frac{1}{\text{E} \left[\left(\frac{\partial \ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)}{\partial \theta} \right)^2 \right]} = -\frac{1}{\text{E} \left[\frac{\partial^2 \ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)}{\partial \theta^2} \right]}$$

- **Minimum Variance Unbiased (MVU):** Estimators that are both unbiased and have the smallest variance of all possible estimators
- Note that these do not necessarily achieve the minimum MSE
- **Cramér-Rao Lower Bound (CRLB)** is a lower bound on unbiased estimators
- Derived in text
- **Log Likelihood Function** of θ is $\ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)$
- Note that the PDF $f_{\mathbf{x};\theta}(\mathbf{x};\theta)$ describes the distribution of the data (stochastic process), not the parameter
- Recall that θ is not a random variable, it is a parameter that defines the distribution

Cramér-Rao Lower Bound Comments

$$\text{var}(\hat{\theta}) \geq \frac{1}{\text{E} \left[\left(\frac{\partial \ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)}{\partial \theta} \right)^2 \right]} = - \frac{1}{\text{E} \left[\frac{\partial^2 \ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)}{\partial \theta^2} \right]}$$

- **Efficient Estimator:** an unbiased estimate that achieves the CRLB with equality
- If it exists, then the unique solution is given by

$$\frac{\partial \ln f_{\mathbf{x};\theta}(\mathbf{x};\theta)}{\partial \theta} = 0$$

where the PDF is evaluated at the observed outcome $\mathbf{x}(\zeta)$

- **Maximum Likelihood (ML) Estimate:** an estimator that satisfies the equation above
- This can be generalized to vectors of parameters
- Limited use — $f_{\mathbf{x};\theta}(\mathbf{x};\theta)$ is rarely known in practice

Consistency

- **Consistent Estimator** an estimator such that

$$\lim_{N \rightarrow \infty} \text{MSE}(\hat{\theta}) = 0$$

- Implies the following as the sample size grows ($N \rightarrow \infty$)
 - The estimator becomes unbiased
 - The variance approaches zero
 - The distribution $f_{\hat{\theta}}(x)$ becomes an impulse centered at θ

Confidence Intervals

- **Confidence Interval:** interval, $a \leq \theta \leq b$, that has a specified probability of covering the unknown true parameter value

$$\Pr \{a < \theta \leq b\} = 1 - \alpha$$

- The interval is estimated from the data, therefore it is also a pair of random variables
- **Confidence Level:** coverage probability of a confidence interval, $1 - \alpha$
- The confidence interval is not uniquely defined by the confidence level
- More later

Properties of the Sample Mean

$$\hat{\mu}_x \triangleq \frac{1}{N} \sum_{n=0}^{N-1} x(n)$$

$$\mathbb{E}[\hat{\mu}_x] = \mu_x$$

$$\text{var}(\hat{\mu}_x) = \frac{1}{N} \sum_{\ell=-N}^N \left(1 - \frac{|\ell|}{N}\right) \gamma_x(\ell) \leq \frac{1}{N} \sum_{\ell=-N}^N \gamma_x(\ell)$$

- If $x(n)$ is WN, then this reduces to $\text{var}(\hat{\mu}_x) = \frac{\sigma_x^2}{N}$
- The estimator is unbiased
- If $\gamma_x(\ell) \rightarrow 0$ as $\ell \rightarrow \infty$, then $\text{var}(\hat{\mu}_x) \rightarrow 0$ (estimator is consistent)
- The variance increases as the correlation of $x(n)$ increases
- In processes with long memory or heavy tails, it is harder to estimate the mean

Sample Mean Confidence Intervals

$$f_{\hat{\mu}_x}(\hat{\mu}_x) = \frac{1}{\sqrt{2\pi}(\sigma_x/\sqrt{N})} \exp \left[-\frac{1}{2} \left(\frac{\hat{\mu}_x - \mu_x}{\sigma_x/\sqrt{N}} \right)^2 \right]$$

$$\Pr \left\{ \mu_x - k \frac{\sigma_x}{\sqrt{N}} < \hat{\mu}_x < \mu_x + k \frac{\sigma_x}{\sqrt{N}} \right\} =$$
$$\Pr \left\{ \hat{\mu}_x - k \frac{\sigma_x}{\sqrt{N}} < \mu_x < \hat{\mu}_x + k \frac{\sigma_x}{\sqrt{N}} \right\} = 1 - \alpha$$

- In general, we don't know the PDF
- If we can assume the process is Gaussian and IID, we know the PDF (sampling distribution) of the estimator
- If N is large and the distribution doesn't have heavy tails, the distribution of $\hat{\mu}_x$ is Gaussian by the Central Limit Theorem (CLT)

Sample Mean Confidence Intervals Comments

$$\Pr \left\{ \hat{\mu}_x - k \frac{\sigma_x}{\sqrt{N}} < \mu_x < \hat{\mu}_x + k \frac{\sigma_x}{\sqrt{N}} \right\} = 1 - \alpha$$

- In many cases the confidence intervals are accurate, even if they are only approximate
- We can choose k such that $1 - \alpha$ equals any probability we like
- In general, the user picks α
- This controls how often the confidence interval does not cover μ_x
- 95% and 99% are common choices

Sample Mean Variance when Gaussian and IID

$$\Pr \left\{ \hat{\mu}_x - k \frac{\sigma_x}{\sqrt{N}} < \mu_x < \hat{\mu}_x + k \frac{\sigma_x}{\sqrt{N}} \right\} = 1 - \alpha$$

- If σ_x is unknown (usually), must estimate from the data

$$\hat{\sigma}_x^2 = \frac{1}{N-1} \sum_{n=0}^{N-1} [x(n) - \hat{\mu}_x]^2$$

- The corresponding z score, has a different distribution
- If $x(n)$ is IID and Gaussian

$$\frac{\hat{\mu}_x - \mu_x}{\hat{\sigma}_x / \sqrt{N}}$$

has a Students' t distribution with $v = N - 1$ *degrees of freedom*

- Approaches a Gaussian distribution as v becomes large (> 20)

Sample Mean Variance when Gaussian

$$\mathbb{E}[\hat{\mu}_x] = \mu_x \quad \text{var}(\hat{\mu}_x) = \frac{1}{N} \sum_{\ell=-N}^N \left(1 - \frac{|\ell|}{N}\right) \gamma_x(\ell)$$

- If $x(n)$ is Gaussian but not IID, the sample mean is normal with mean μ
- The *approximate* confidence interval is given by a Gaussian PDF

$$\Pr \left\{ \hat{\mu}_x - k \sqrt{\text{var}(\hat{\mu}_x)} < \mu_x < \hat{\mu}_x + k \sqrt{\text{var}(\hat{\mu}_x)} \right\} = 1 - \alpha$$

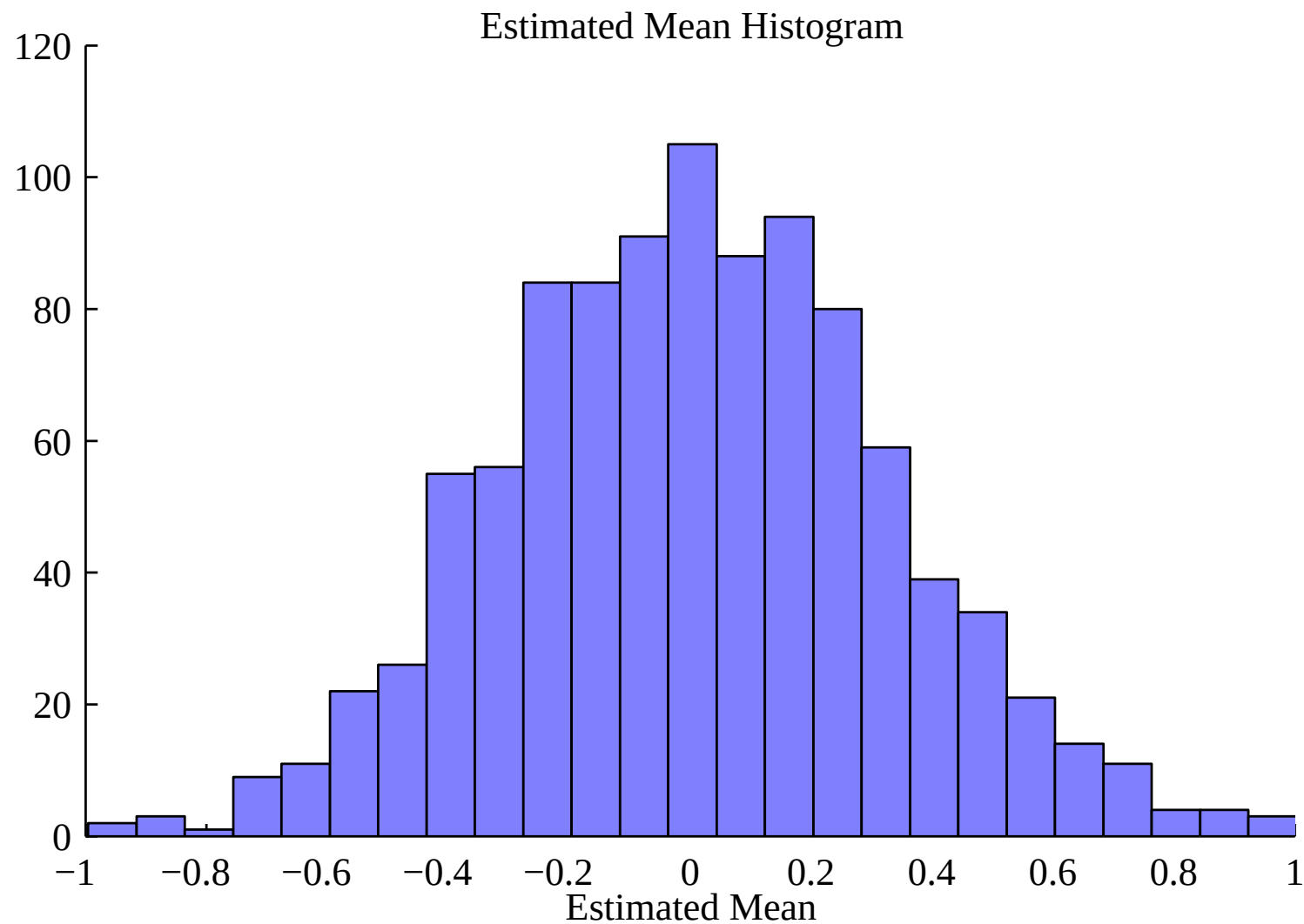
- Note that $\text{var}(\hat{\mu}_x)$ requires knowledge $\gamma_x(\ell)$

Example 1: Mean Confidence Intervals

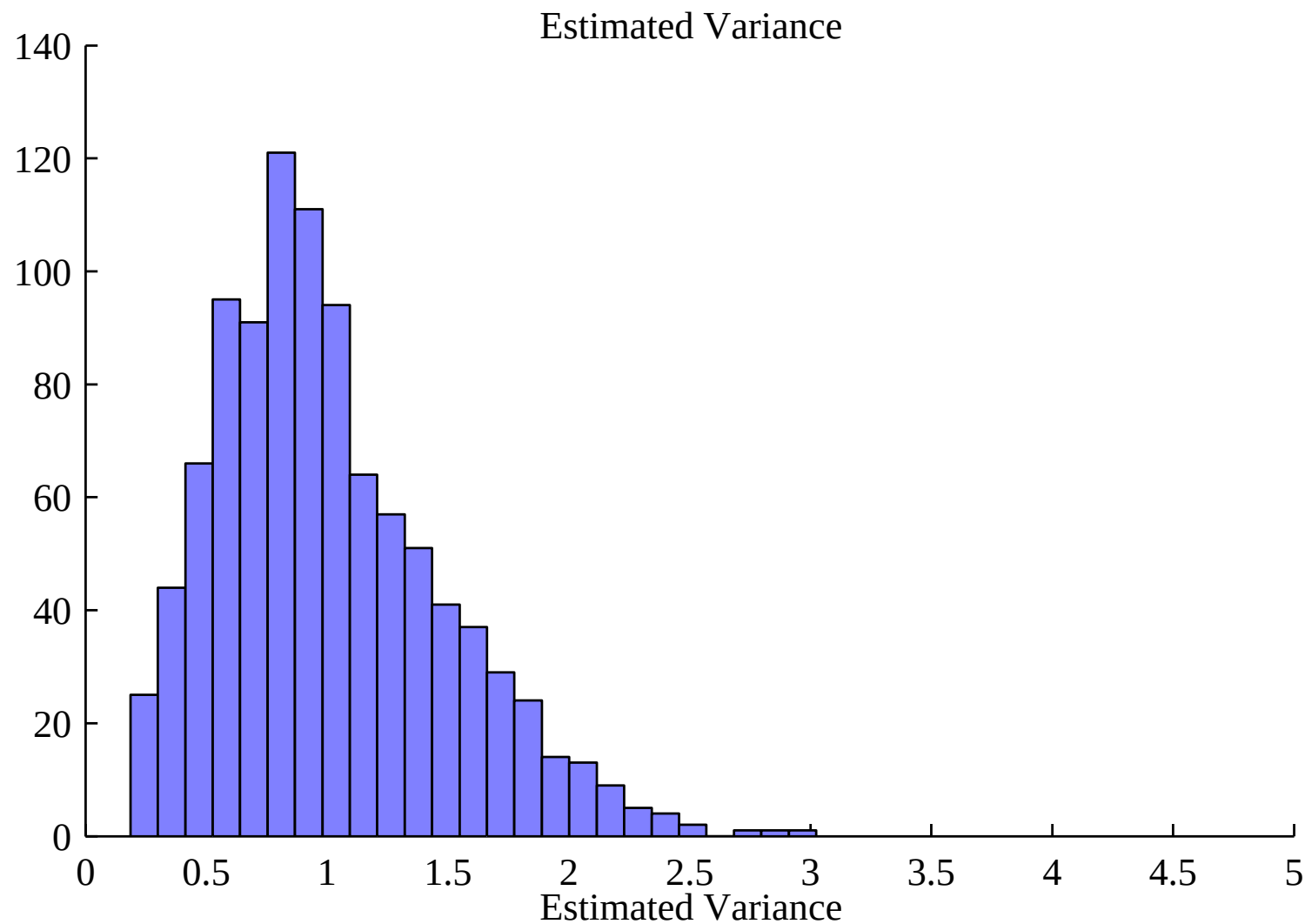
Generate 1000 random experiments of a white noise signal of length $N = 10$ and $N = 100$. Plot the histograms of the 95% confidence intervals, the means, and specify the percentage of times that the true mean was within the confidence interval. Repeat for a Gaussian and Exponential distributions.

- $N = 10$, Normal: 94.4% Coverage
- $N = 10$, Exponential: 88.9% Coverage
- $N = 100$, Normal: 95.7% Coverage
- $N = 100$, Exponential: 95.1% Coverage

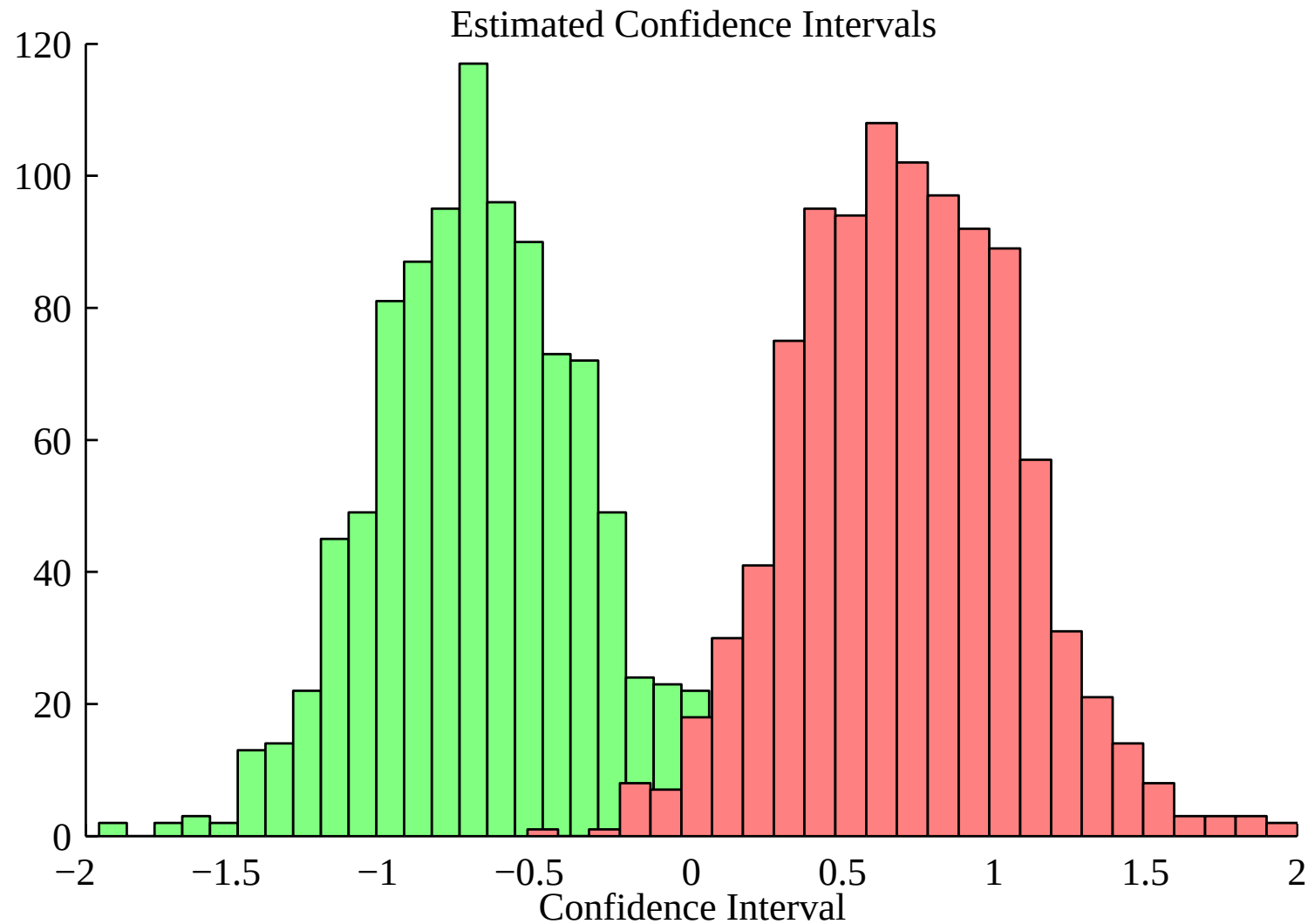
Example 1: Mean Histogram, $N = 10$



Example 1: Variance Histogram, $N = 10$



Example 1: Confidence Interval Histogram, $N = 10$



Example 1: MATLAB Code

```
%function [] = TransformDomain();
close all;
clear all;

M = 1000; % No. experiments
N = 100;  % No. observations
cl = 95;  % Confidence level

%ds = 'Exponential';
%tm = 1;    % True mean
%X = exprnd(tm,N,M);

ds = 'Normal';
tm = 0;    % True mean
X = randn(N,M);

%X = sqrt(12)*(rand(N,M)-0.5);

mx = mean(X); % Estimate the mean
sx = std(X);  % Uses the correct equation for std. dev. estimation
lc = mx + sx*tinv( (1-cl/100)/2,N-1)/sqrt(N); % Lower confidence interval
uc = mx + sx*tinv(1-(1-cl/100)/2,N-1)/sqrt(N); % Upper confidence interval

fprintf('Mean covered: %5.2f%c\n',100*sum(lc<tm & uc>=tm)/M,char(37));

%=====
% Plot the Mean Histogram
```

```

%=====
figure
FigureSet(1,4.5,2.8);
[n,x] = hist(mx,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[0.5 0.5 1.0]);
xlim(tm+[-1 1]);
title('Estimated Mean Histogram');
xlabel('Estimated Mean');
%xlim([min(n) max(n)]);
box off;
AxisSet(8);
eval(sprintf('print -depsc %sMeanHistogram%03d;',ds,N));

%=====
% Plot the Standard Deviation Histogram
%=====
figure
FigureSet(1,4.5,2.8);
[n,x] = hist(sx.^2,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[0.5 0.5 1.0]);
xlim([0 5]);
title('Estimated Variance');
xlabel('Estimated Variance');
box off;
AxisSet(8);
eval(sprintf('print -depsc %sVarianceHistogram%03d;',ds,N));

%=====
% Plot the Confidence intervals

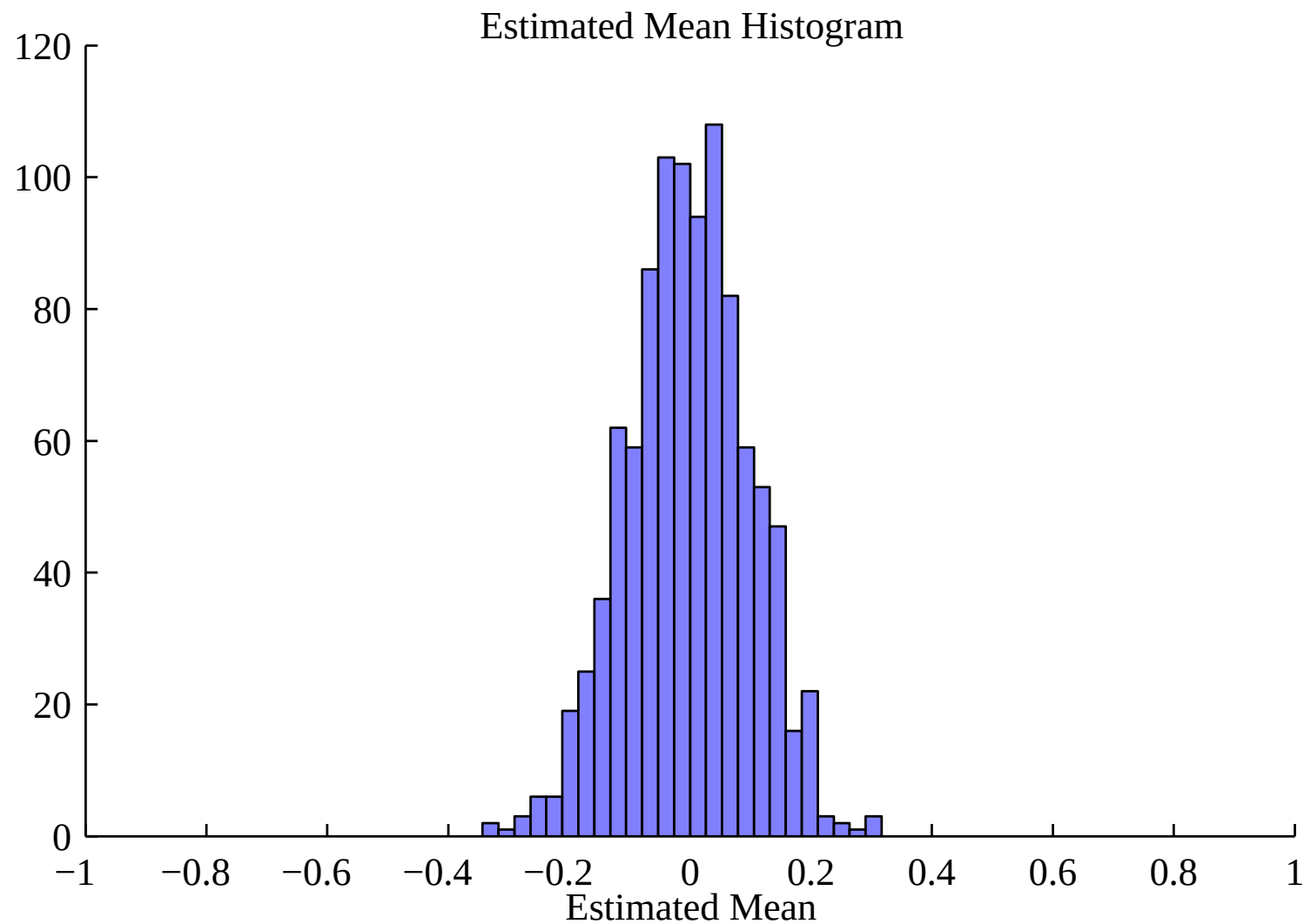
```

```

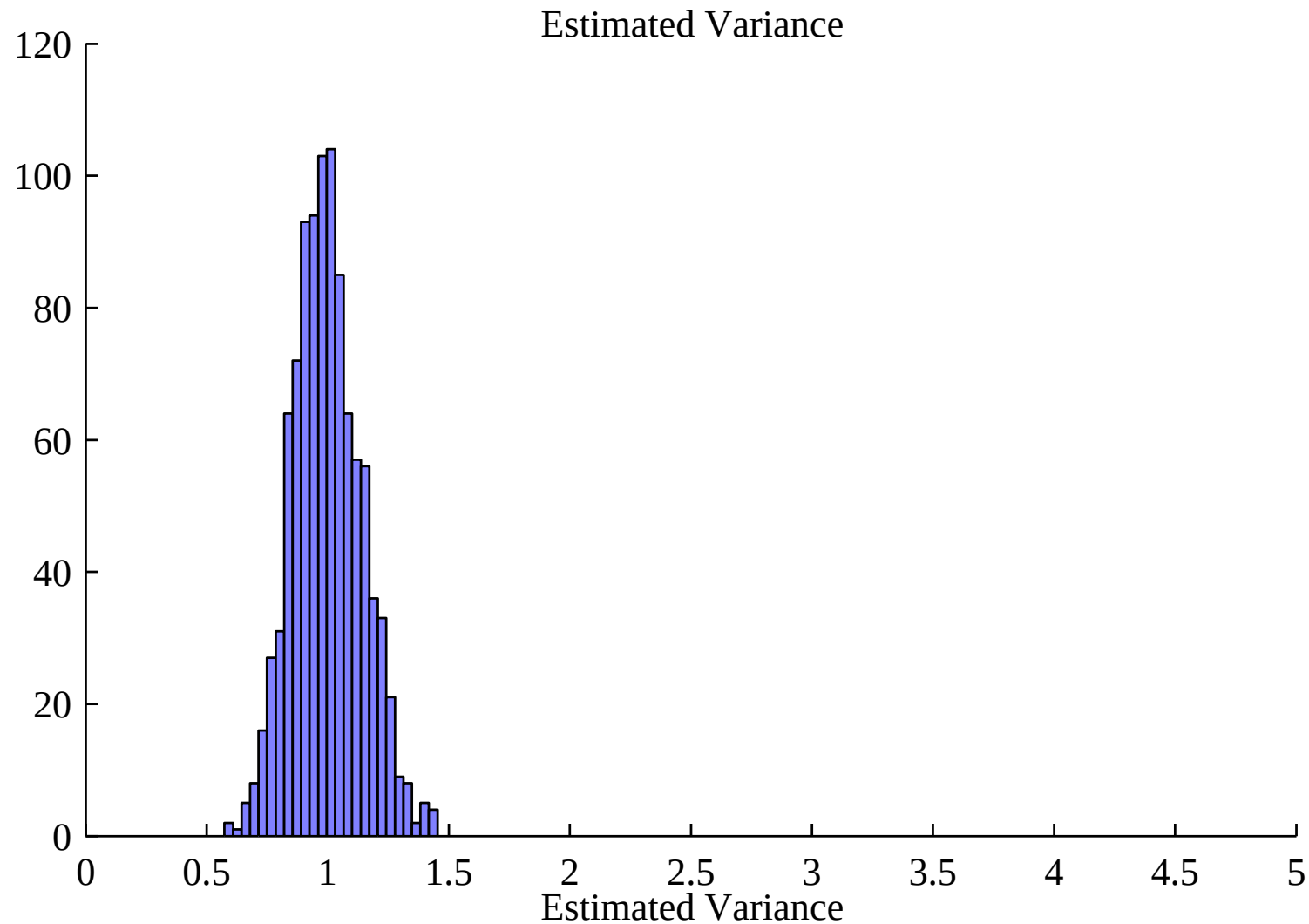
%=====
figure
FigureSet(1,4.5,2.8);
[n,x] = hist(uc,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[0.5 1.0 0.5]);
hold on;
    [n,x] = hist(lc,25);
    h = bar(x,n,1.0);
    set(h,'FaceColor',[1.0 0.5 0.5]);
    hold off;
xlim(tm+[-2 2]);
title('Estimated Confidence Intervals');
xlabel('Confidence Interval');
box off;
AxisSet(8);
eval(sprintf('print -depsc %sConfidenceHistogram%03d;',ds,N));

```

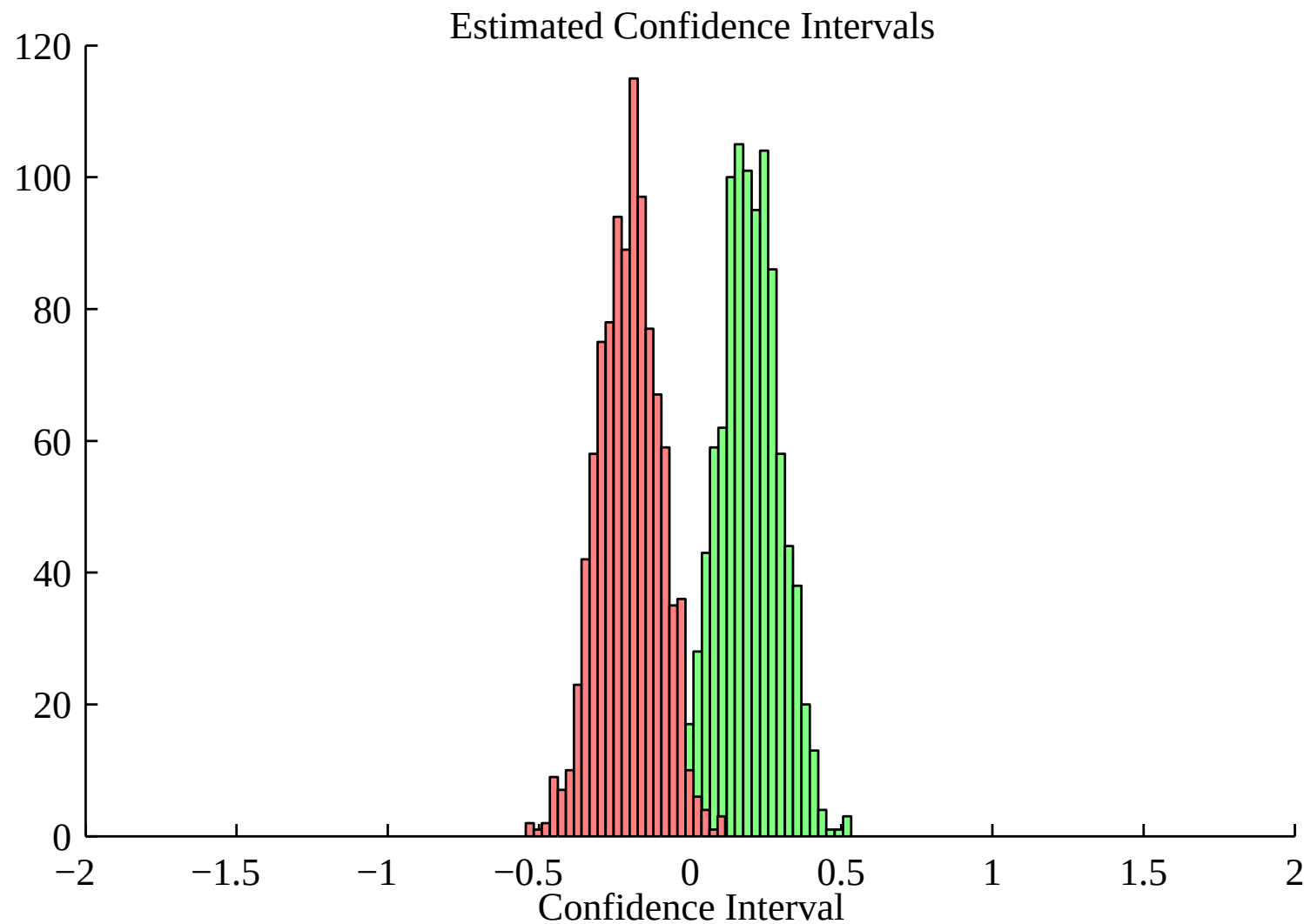
Example 1: Mean Histogram, Normal $N = 100$



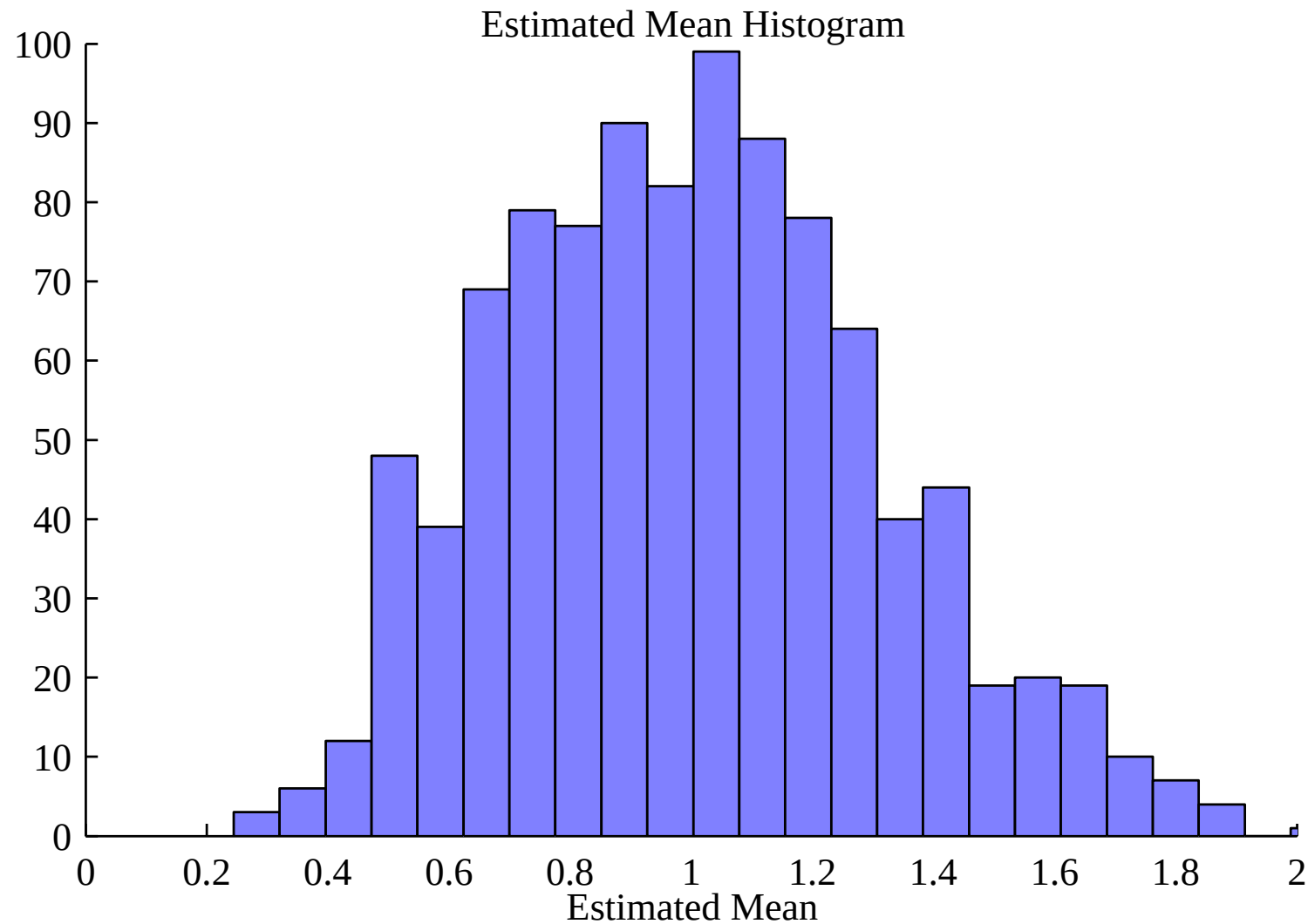
Example 1: Variance Histogram, Normal $N = 100$



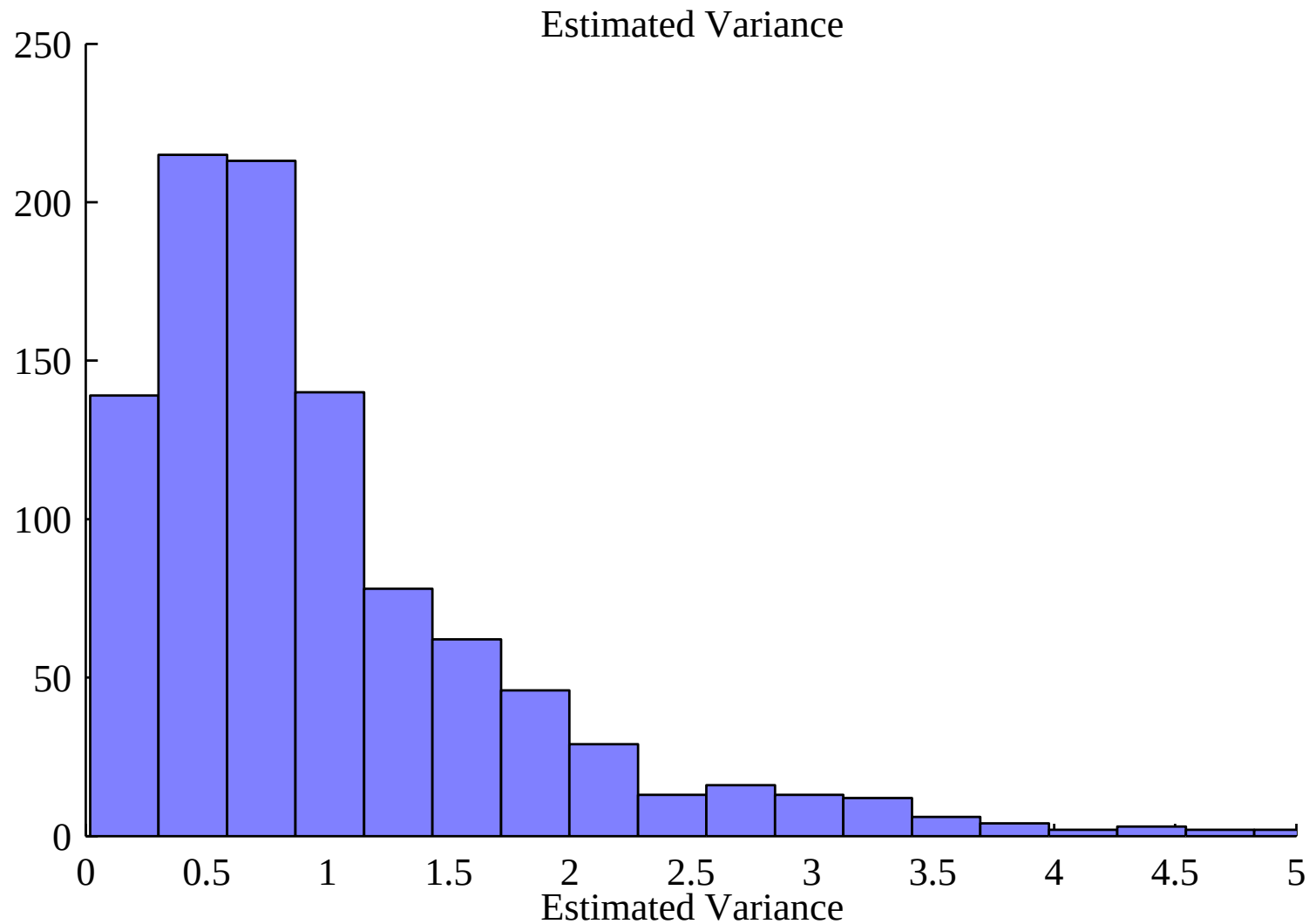
Example 1: Confidence Interval Histogram, Normal $N = 100$



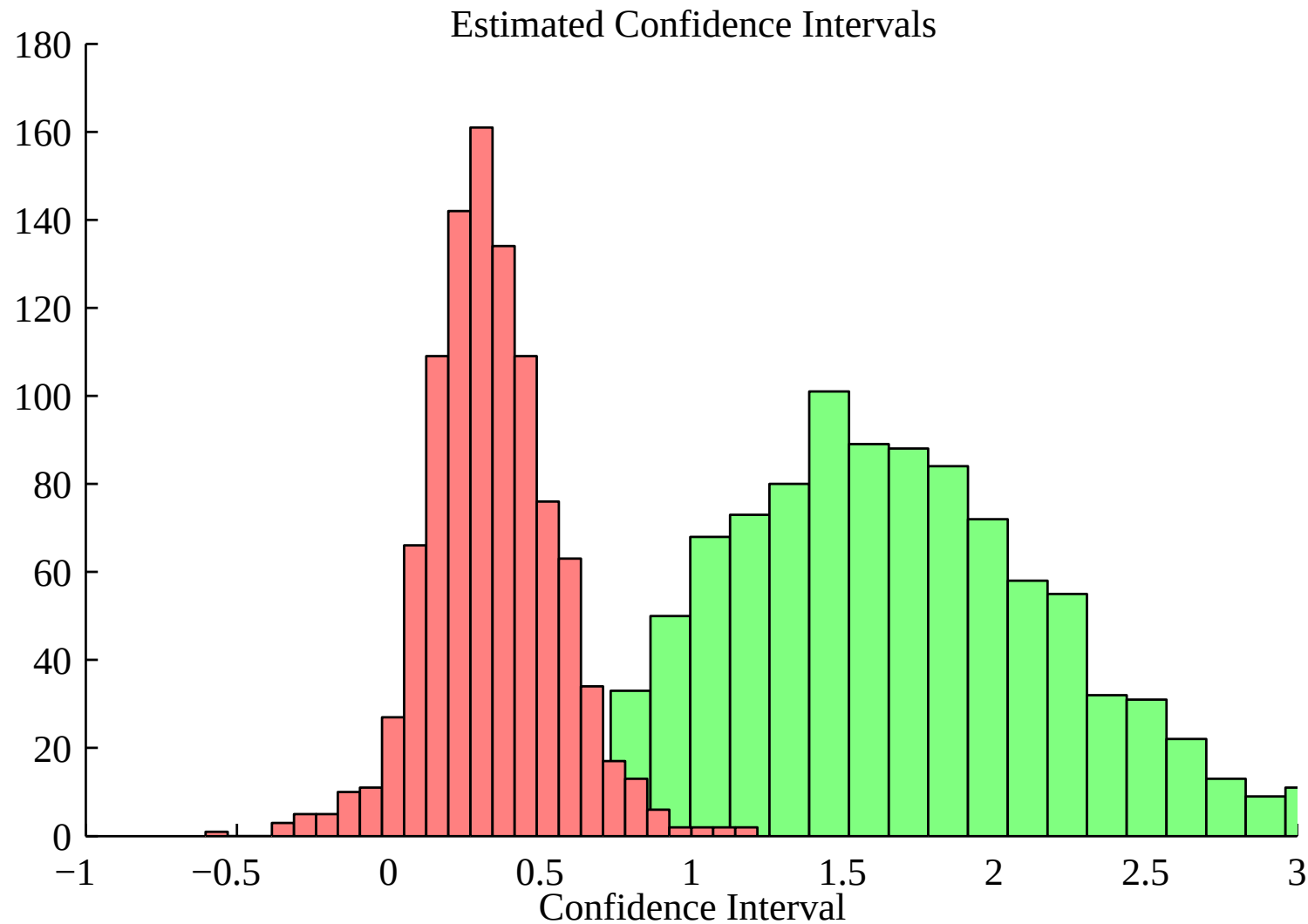
Example 1: Mean Histogram, Exponential $N = 10$



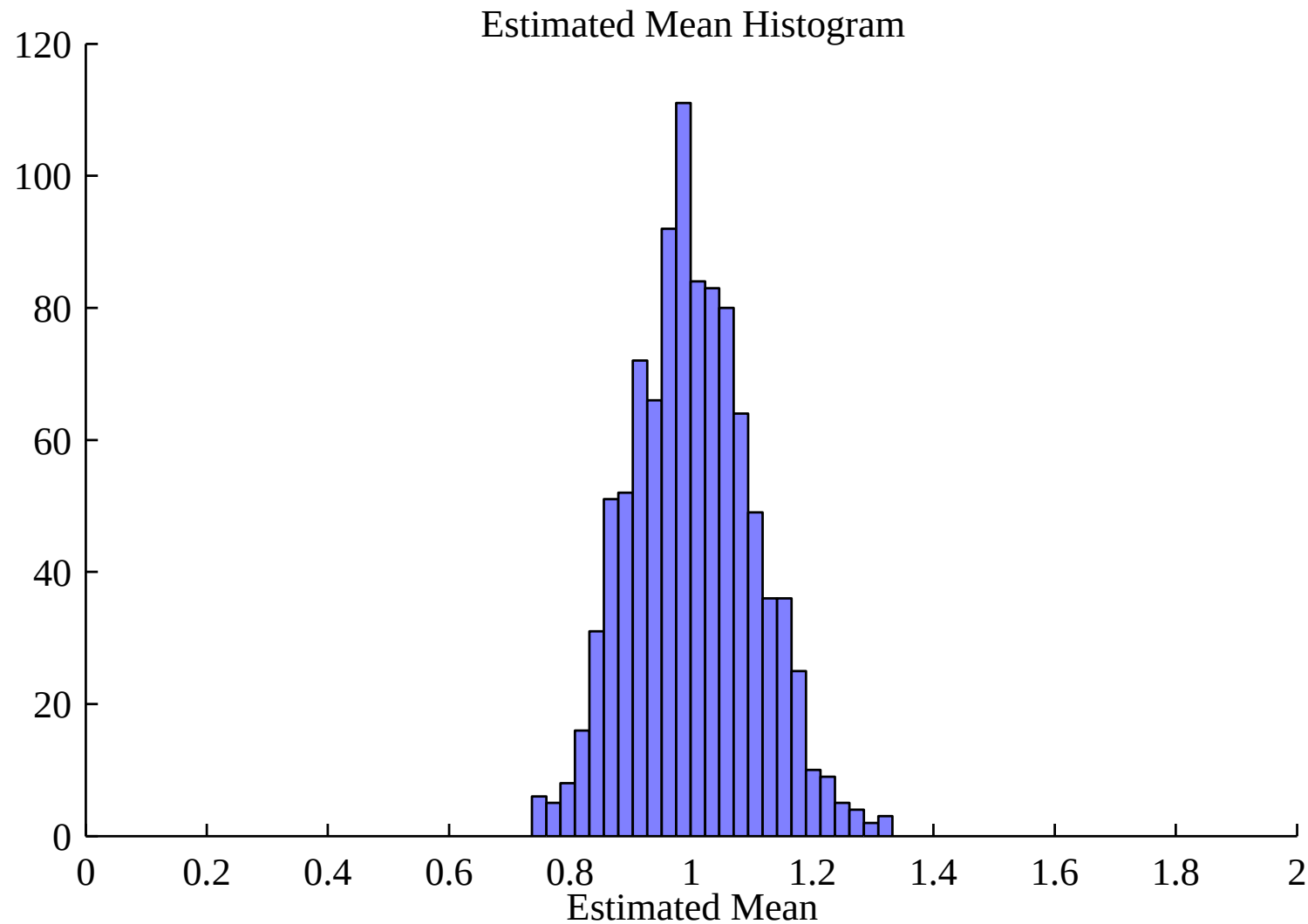
Example 1: Variance Histogram, Exponential $N = 10$



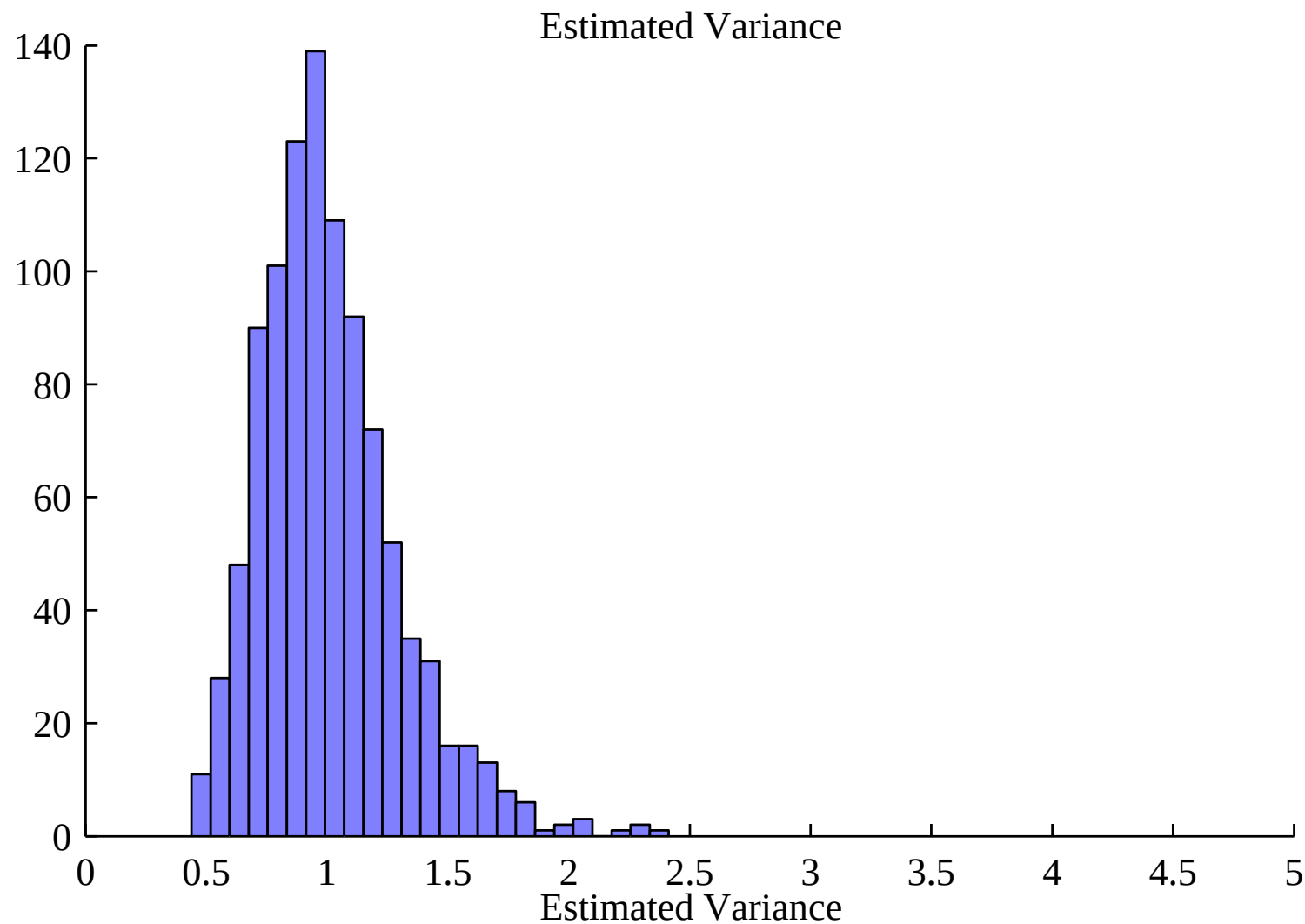
Example 1: Confidence Interval Histogram, Exponential $N = 10$



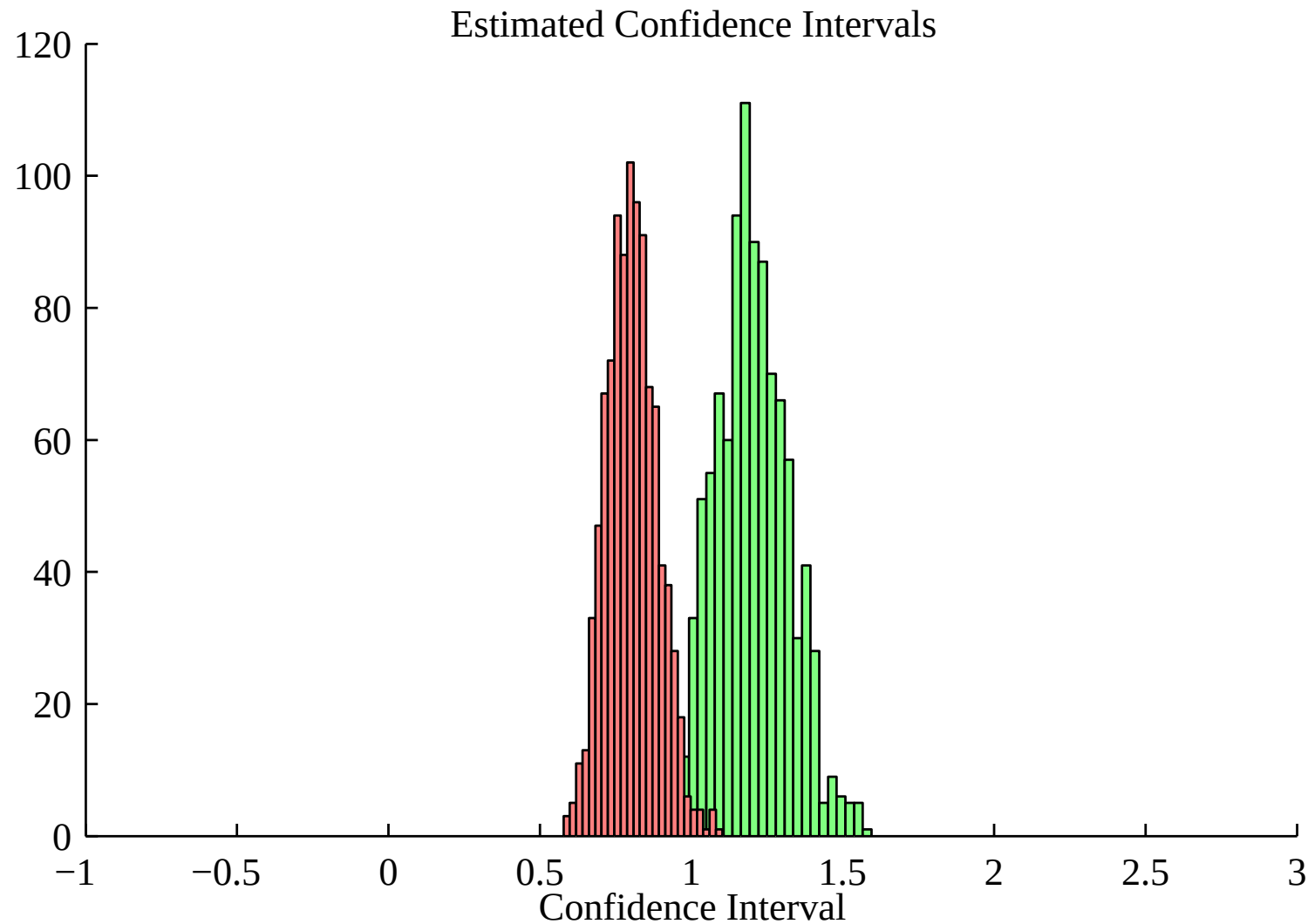
Example 1: Mean Histogram, Exponential $N = 100$



Example 1: Variance Histogram, Exponential $N = 100$



Example 1: Confidence Interval Histogram, Exponential $N = 100$



Estimation of Variance

The “natural estimator” of the variance is

$$\hat{\sigma}_x^2 \triangleq \frac{1}{N} \sum_{n=0}^{N-1} [x(n) - \hat{\mu}_x]^2$$

In general, the mean is given by

$$\mathbb{E}[\hat{\sigma}_x^2] = \sigma_x^2 - \text{var}(\hat{\mu}_x) = \sigma_x^2 - \frac{1}{N} \sum_{\ell=-N}^N \left(1 - \frac{|\ell|}{N}\right) \gamma_x(\ell)$$

If $x(n)$ is uncorrelated this reduces to

$$\mathbb{E}[\hat{\sigma}_x^2] = \frac{N-1}{N} \sigma_x^2$$

Thus, $\hat{\sigma}_x^2$ is a biased estimator!

Example 2: Biased Variance

Let $w(n) \sim \text{WN}(0, \sigma_w^2)$. Find a closed-form expression for $E[\hat{\sigma}_w^2]$ where $\hat{\sigma}_w^2$ is the natural variance estimator in terms of σ_w^2 and the length of the sequence N .

Example 2: Workspace

Estimation of Variance

A better estimator (if the mean is unknown) is

$$\hat{\sigma}_x^2 \triangleq \frac{1}{N-1} \sum_{n=0}^{N-1} [x(n) - \hat{\mu}_x]^2$$

$$\text{var}(\hat{\sigma}_x^2) \approx \frac{2\sigma^4}{N-1} \text{ for large } N$$

- If $x(n)$ is uncorrelated, this estimator is unbiased
- As $N \rightarrow \infty$, if $\gamma_x(\ell) \rightarrow 0$ as $\ell \rightarrow \infty$, then $\text{var}(\hat{\sigma}_x^2) \rightarrow 0$ and the biased estimator is *asymptotically unbiased*
- Both estimators are consistent

Sample Variance Confidence Intervals

$$\hat{\sigma}_x^2 \triangleq \frac{1}{N-1} \sum_{n=0}^{N-1} [x(n) - \hat{\mu}_x]^2$$

- If the samples are IID and Gaussian, $(N-1)\hat{\sigma}_x^2/\sigma_x^2$ has a *chi-squared* distribution with N degrees of freedom

$$\Pr \left\{ \hat{\sigma}_x^2 \frac{N-1}{\chi_v(0.975)} < \sigma_x^2 \leq \hat{\sigma}_x^2 \frac{N-1}{\chi_v(0.025)} \right\} = 1 - \alpha$$

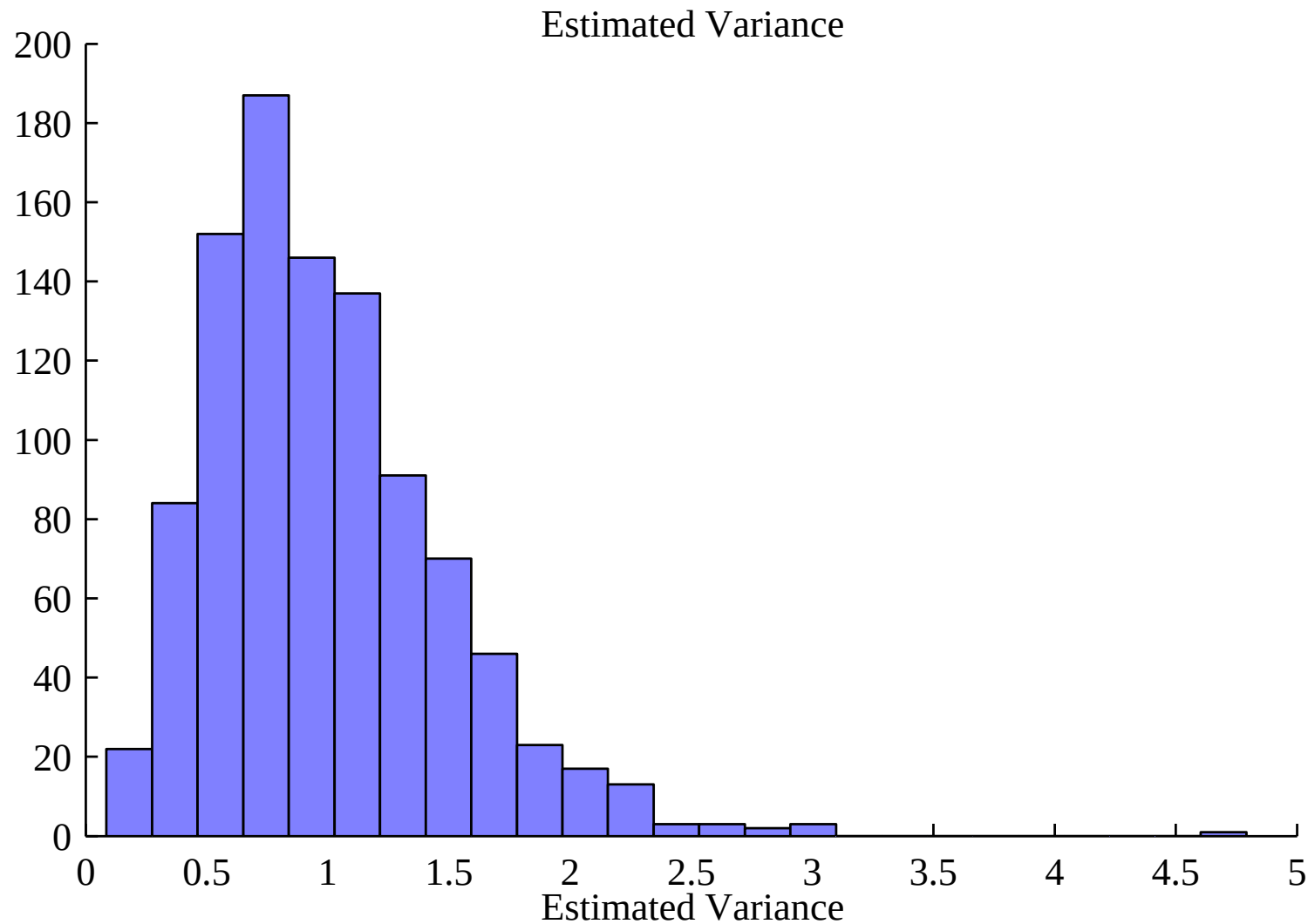
- The quantiles of $\chi_v(\cdot)$ can be obtained from look-up tables or MATLAB
- This confidence interval is sensitive to the normal assumption (unlike the confidence intervals for the mean)
- Also sensitive to the IID assumption (like the mean)

Example 3: Variance Confidence Intervals

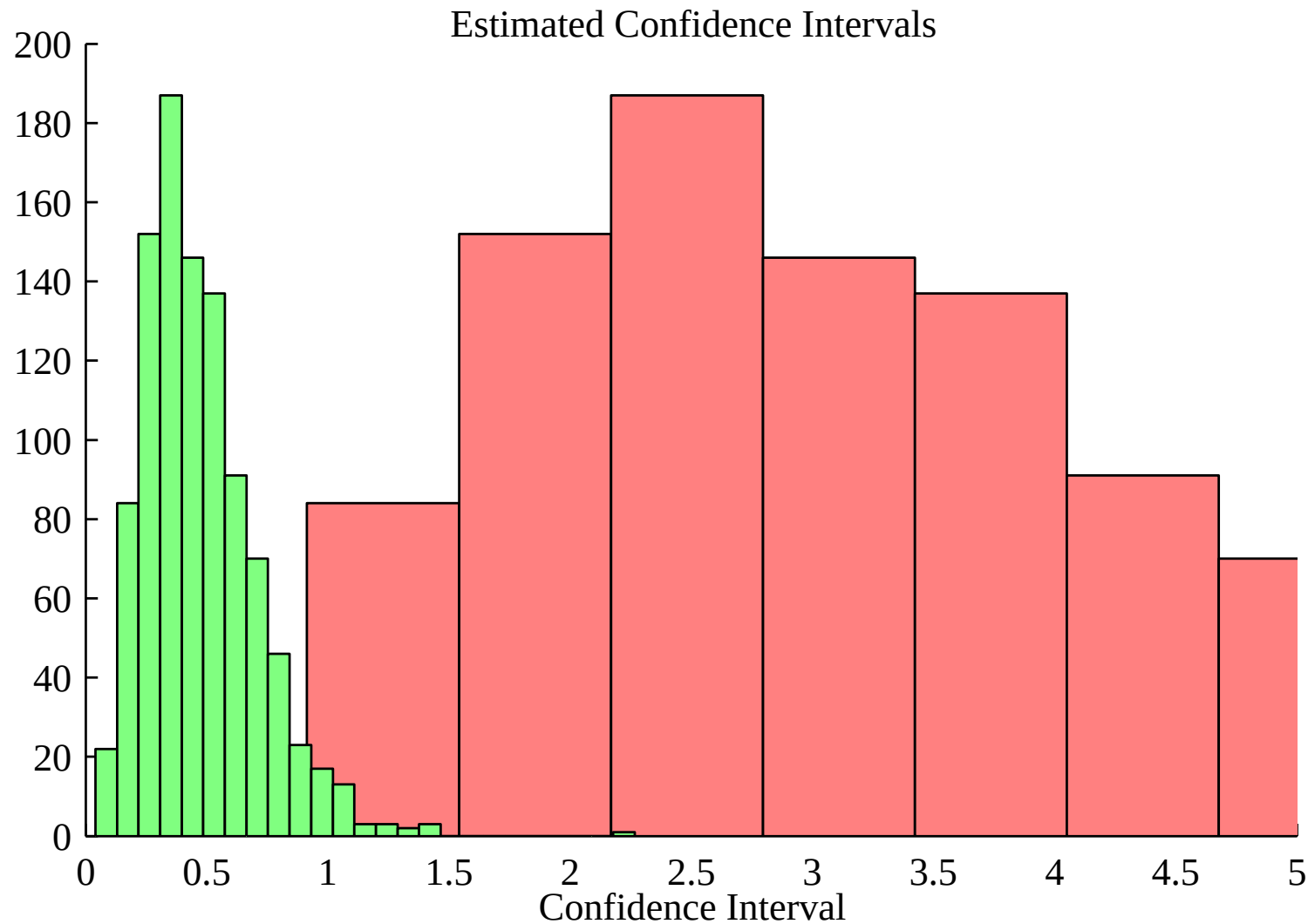
Generate 1000 random experiments of a white noise signal of length $N = 10$ and $N = 100$. Plot the histograms of the estimated variances, 95% confidence intervals, and the confidence interval lengths. Specify the percentage of times that the true variance was within the confidence interval. Repeat for a Gaussian and Exponential distributions.

- $N = 10$, Normal: 94.9% Coverage
- $N = 10$, Exponential: 76.0% Coverage
- $N = 100$, Normal: 95.6% Coverage
- $N = 100$, Exponential: 68.4% Coverage

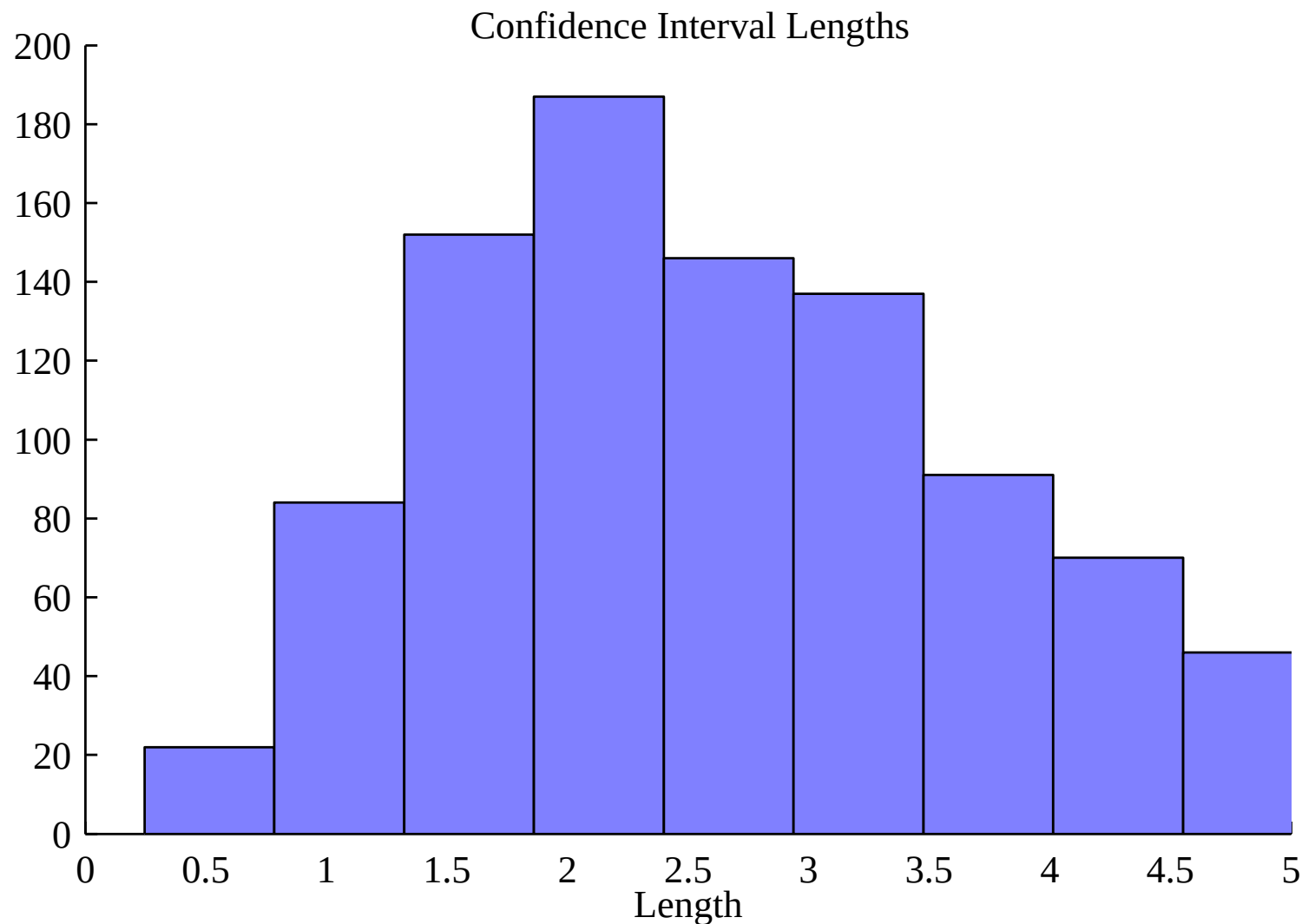
Example 3: Variance Histogram, Normal $N = 10$



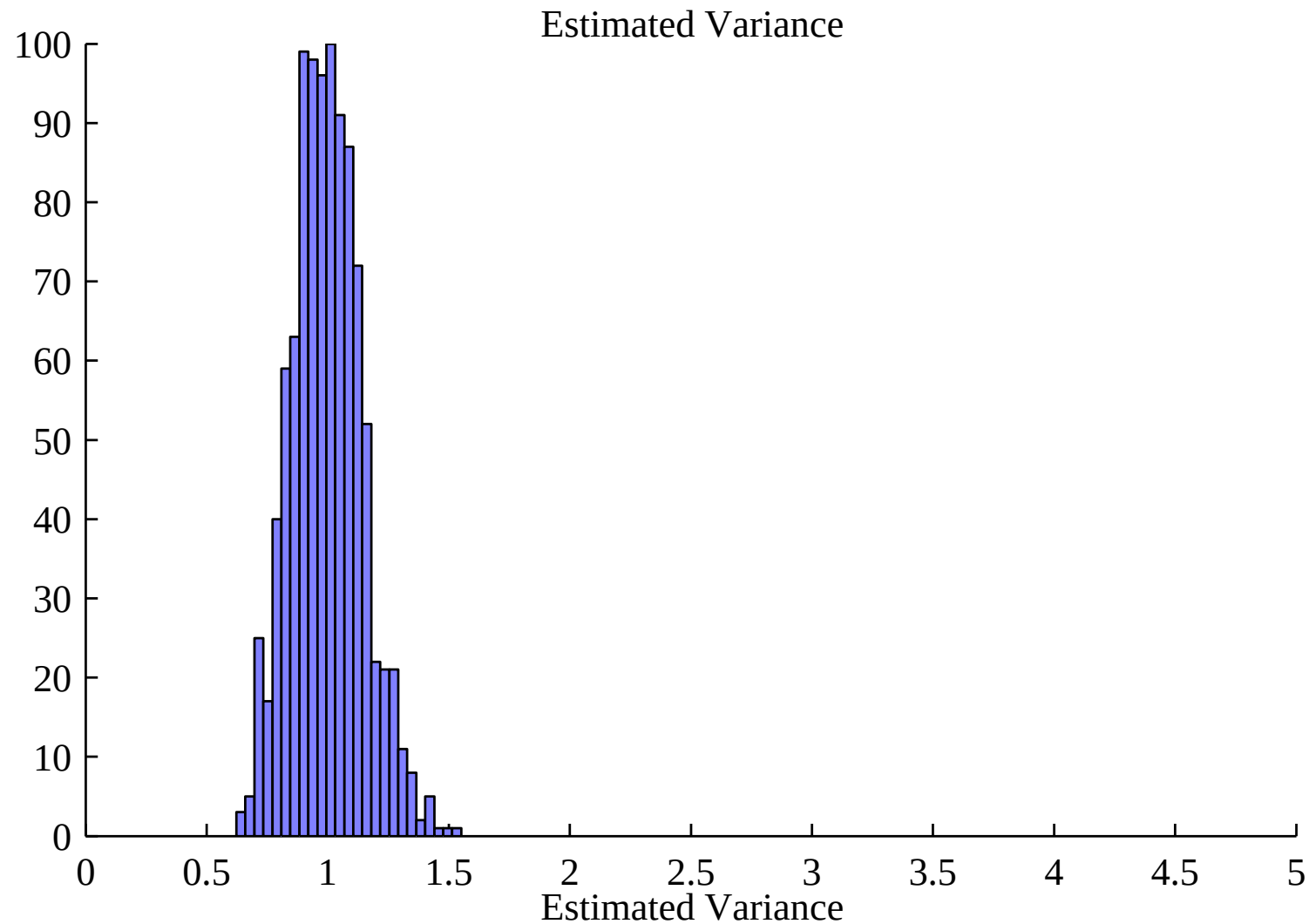
Example 3: Confidence Interval Histogram, Normal $N = 10$



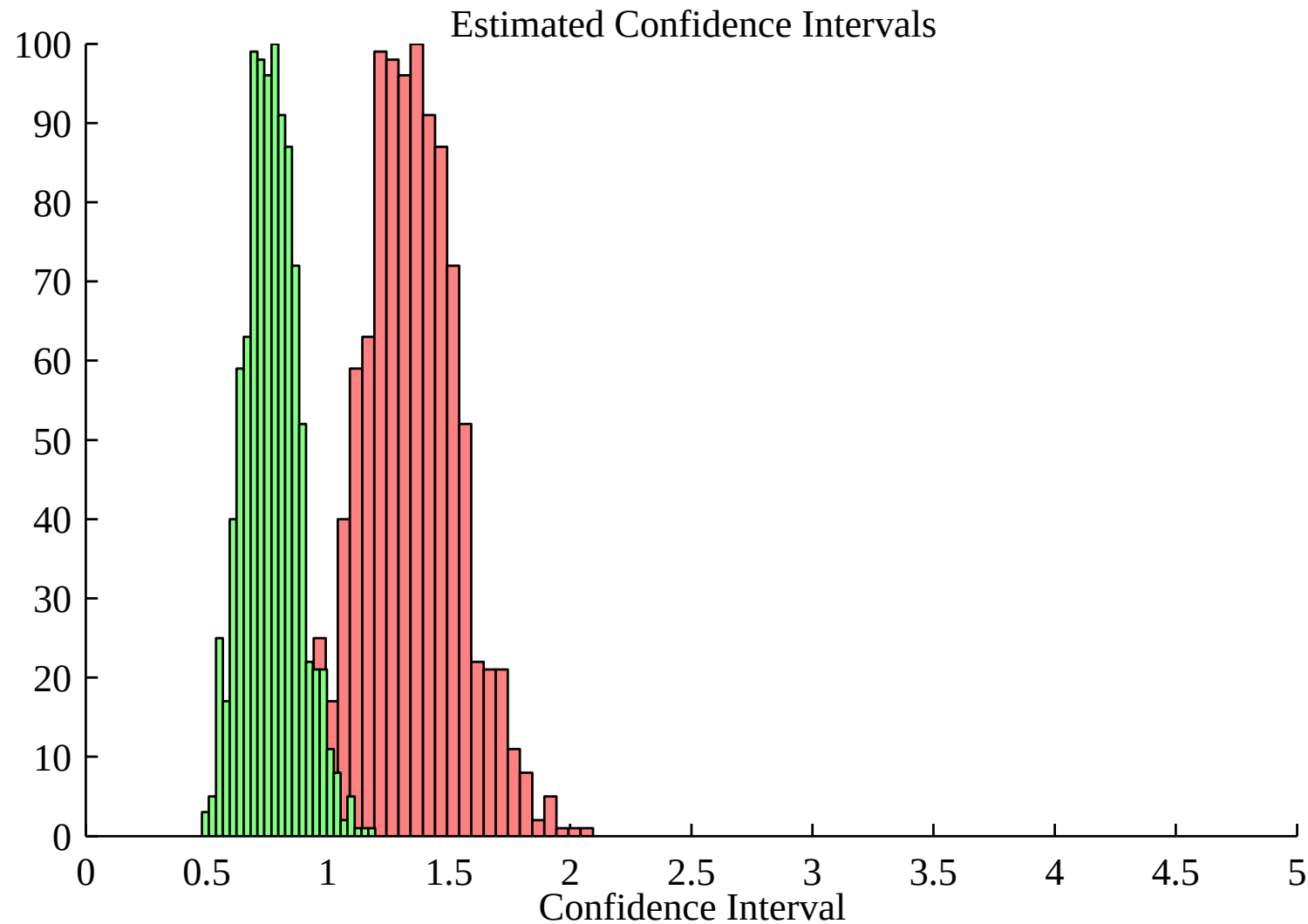
Example 3: Confidence Length Histogram, Normal $N = 10$



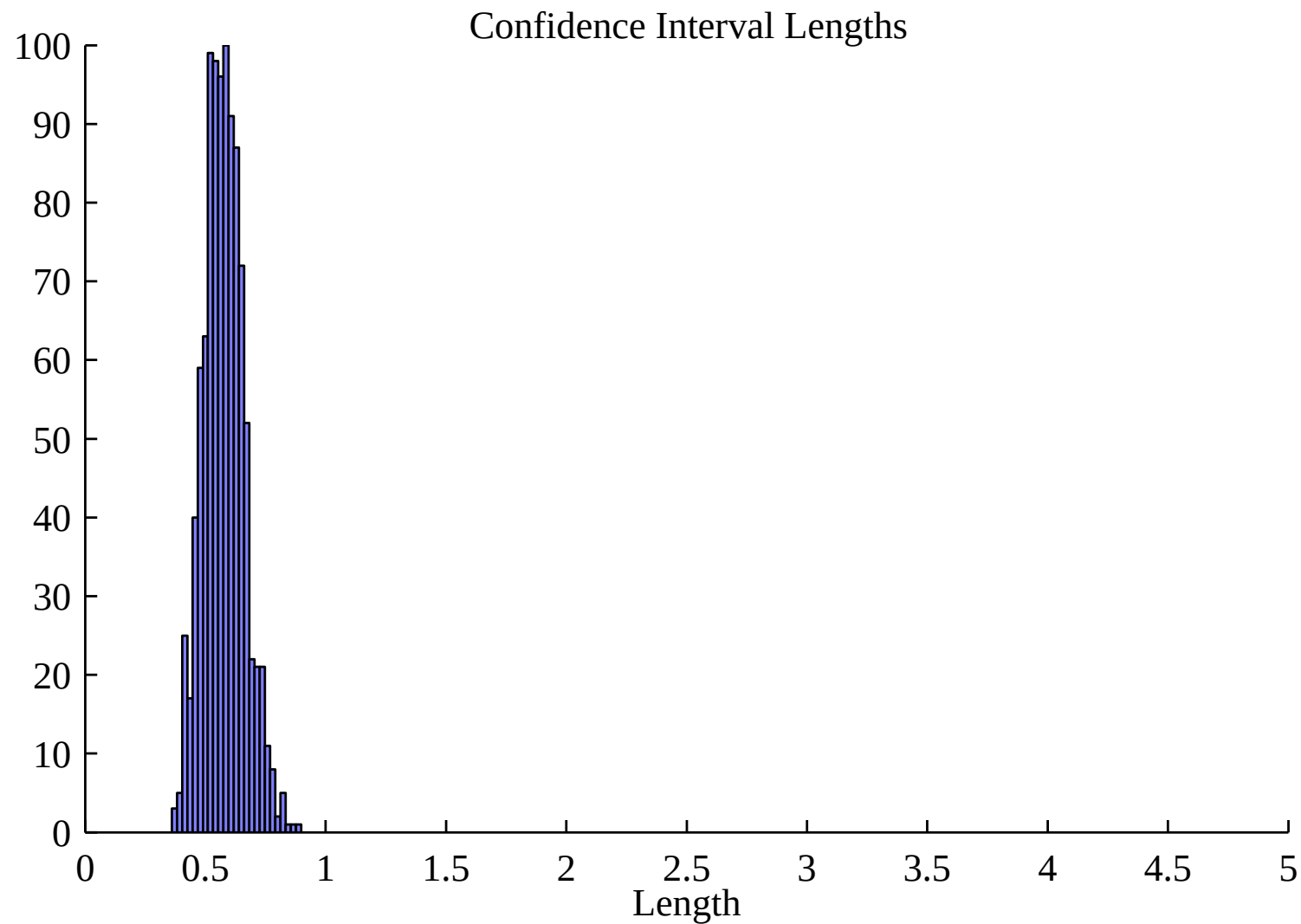
Example 3: Variance Histogram, Normal $N = 100$



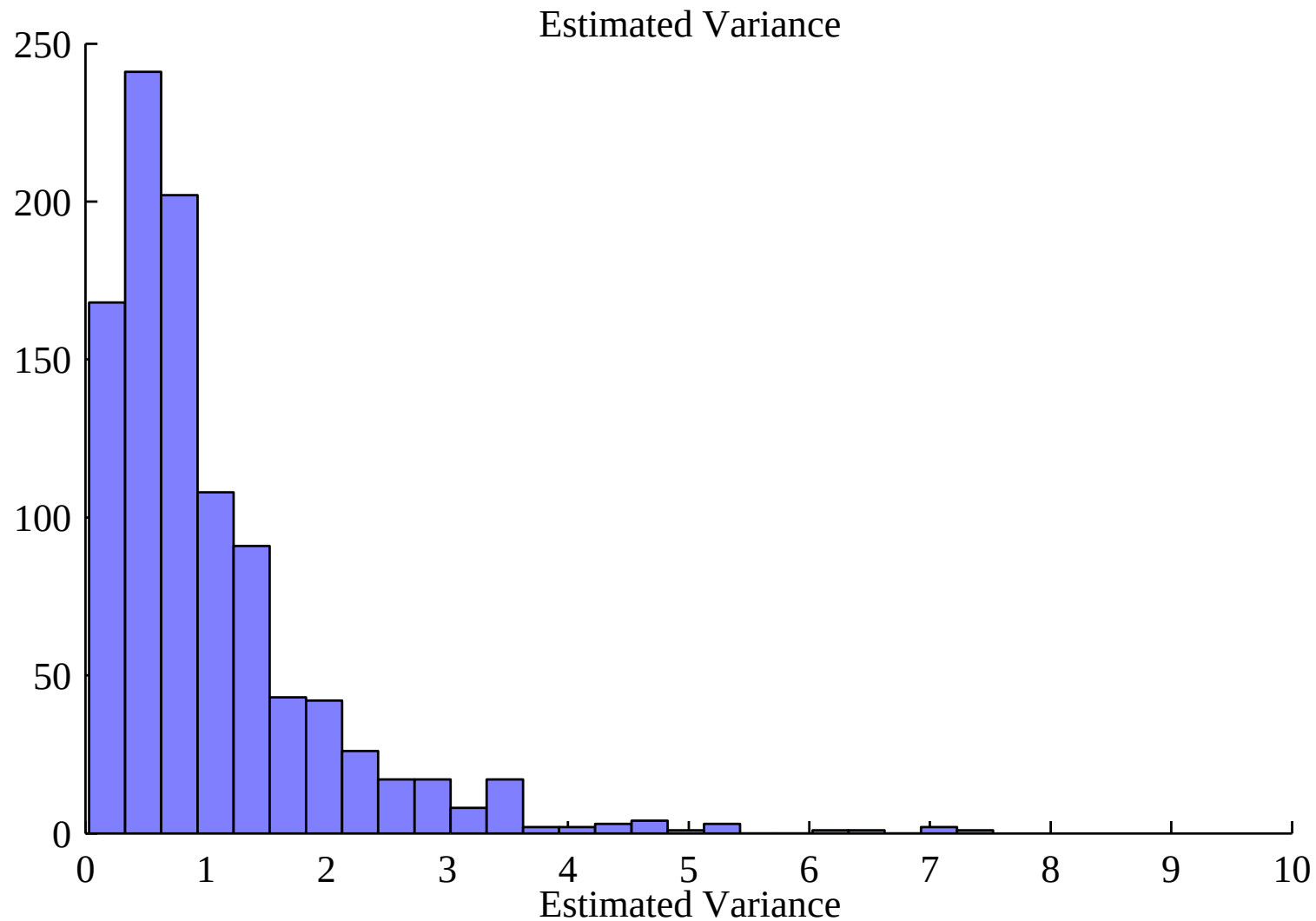
Example 3: Confidence Interval Histogram, Normal $N = 100$



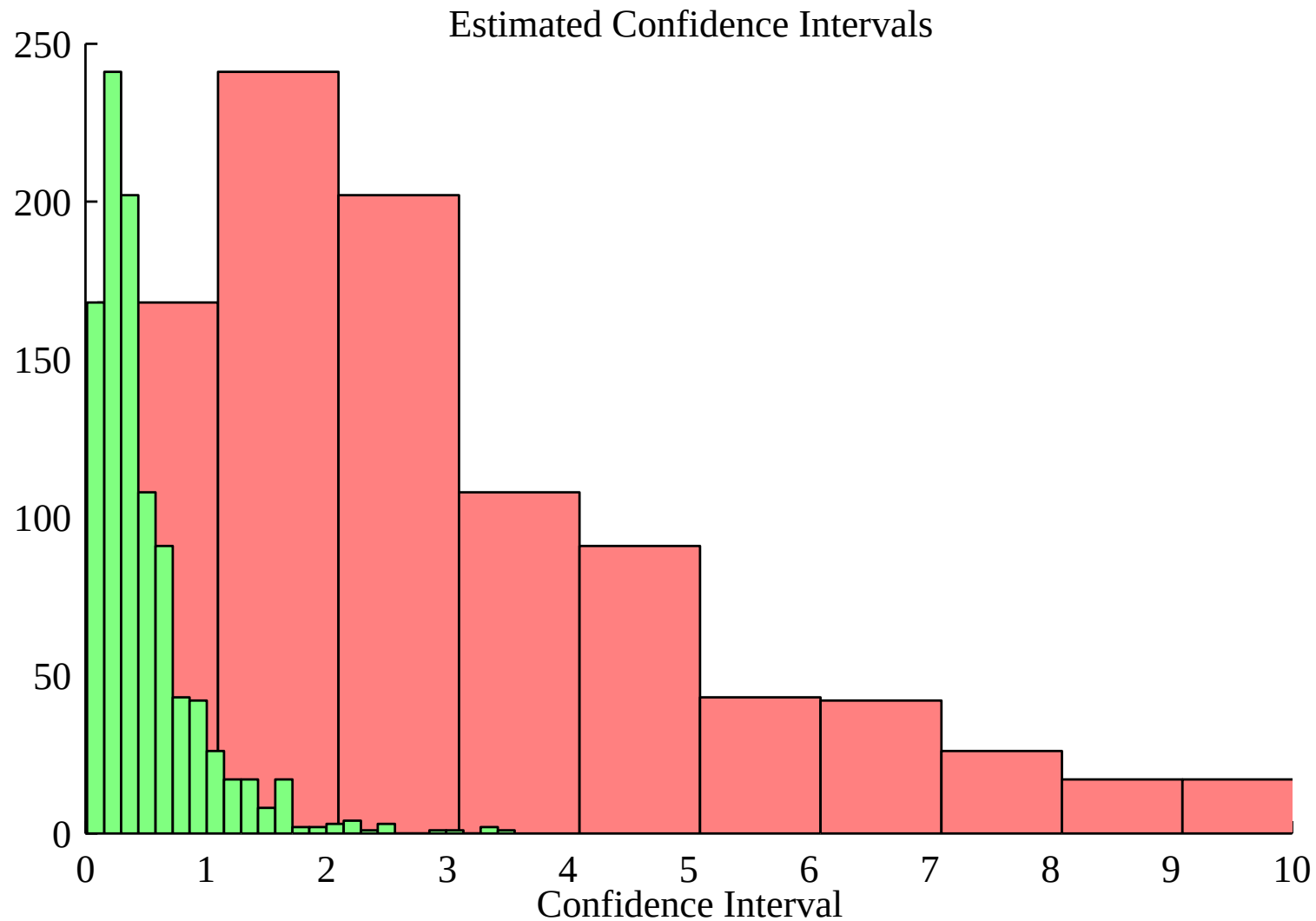
Example 3: Confidence Length Histogram, Normal $N = 100$



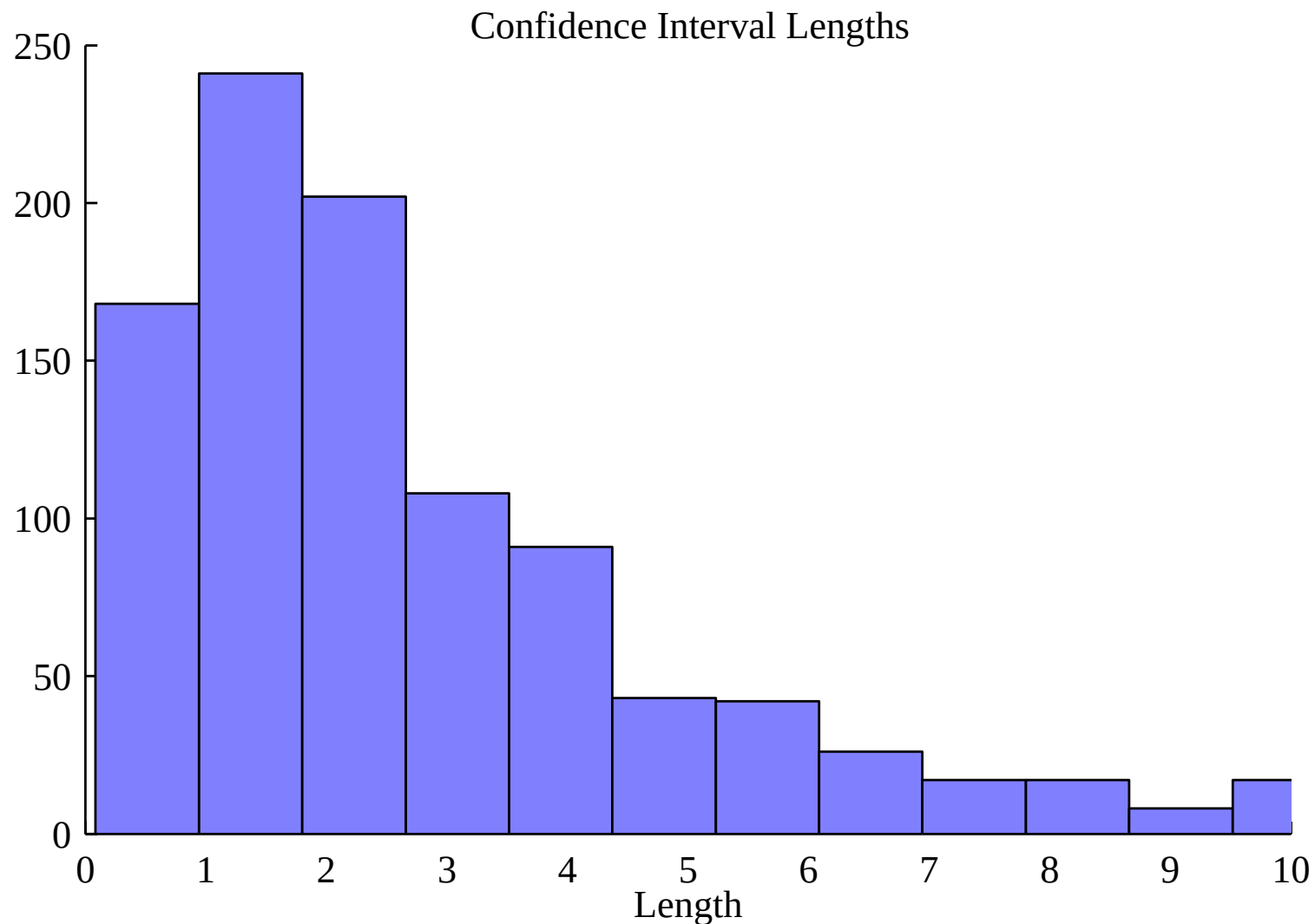
Example 3: Variance Histogram, Exponential $N = 10$



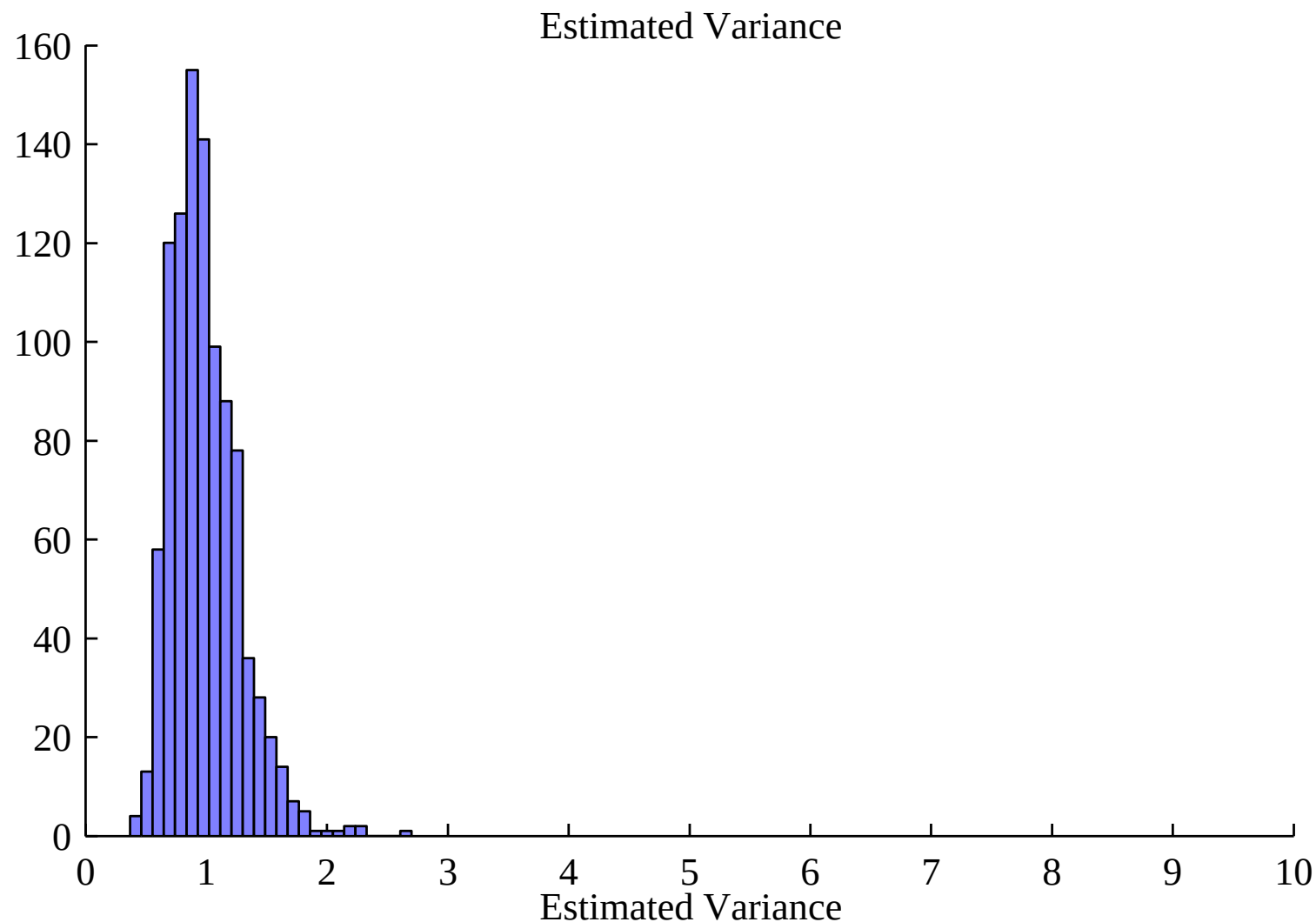
Example 3: Confidence Interval Histogram, Exponential $N = 10$



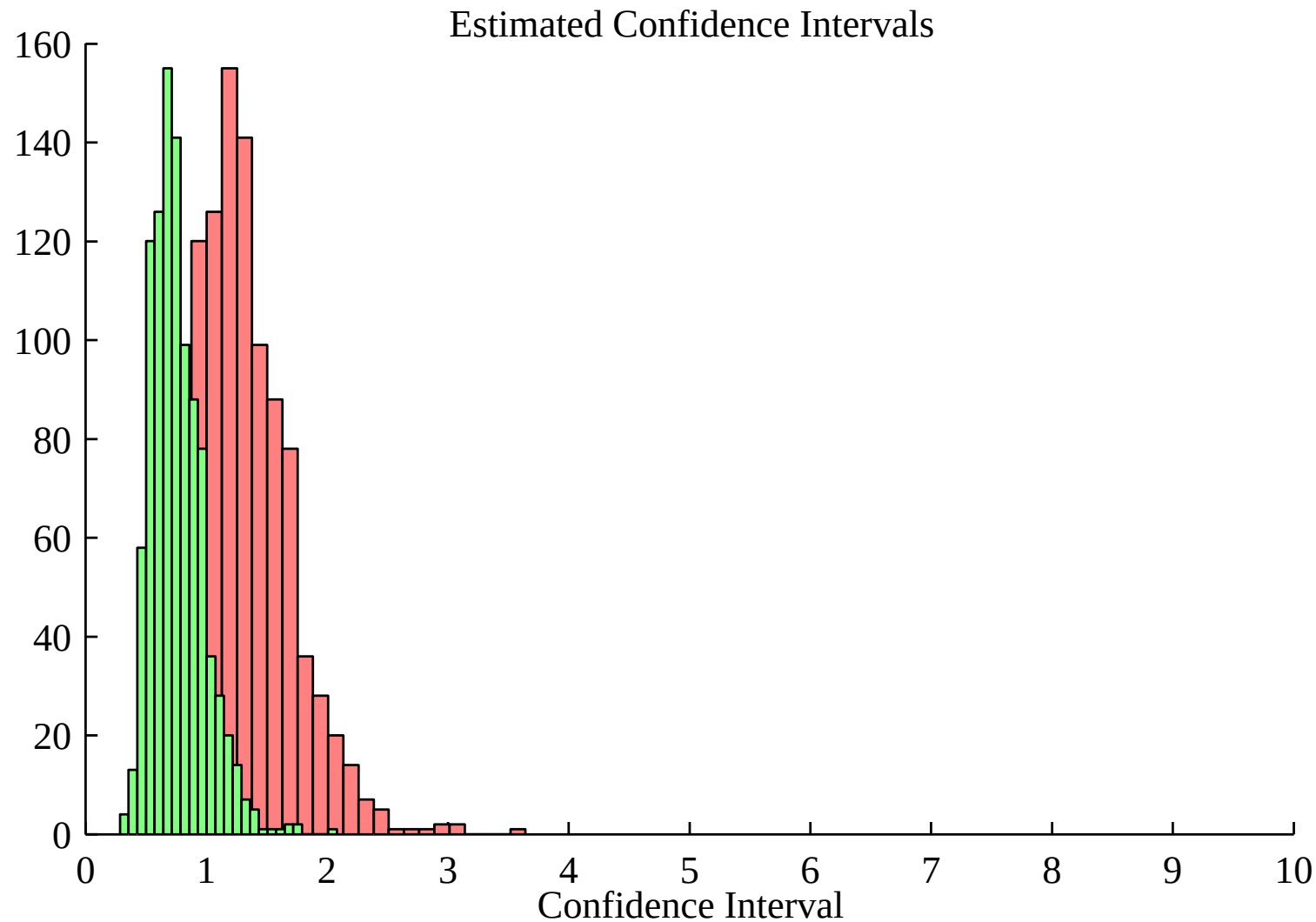
Example 3: Confidence Length Histogram, Exponential $N = 10$



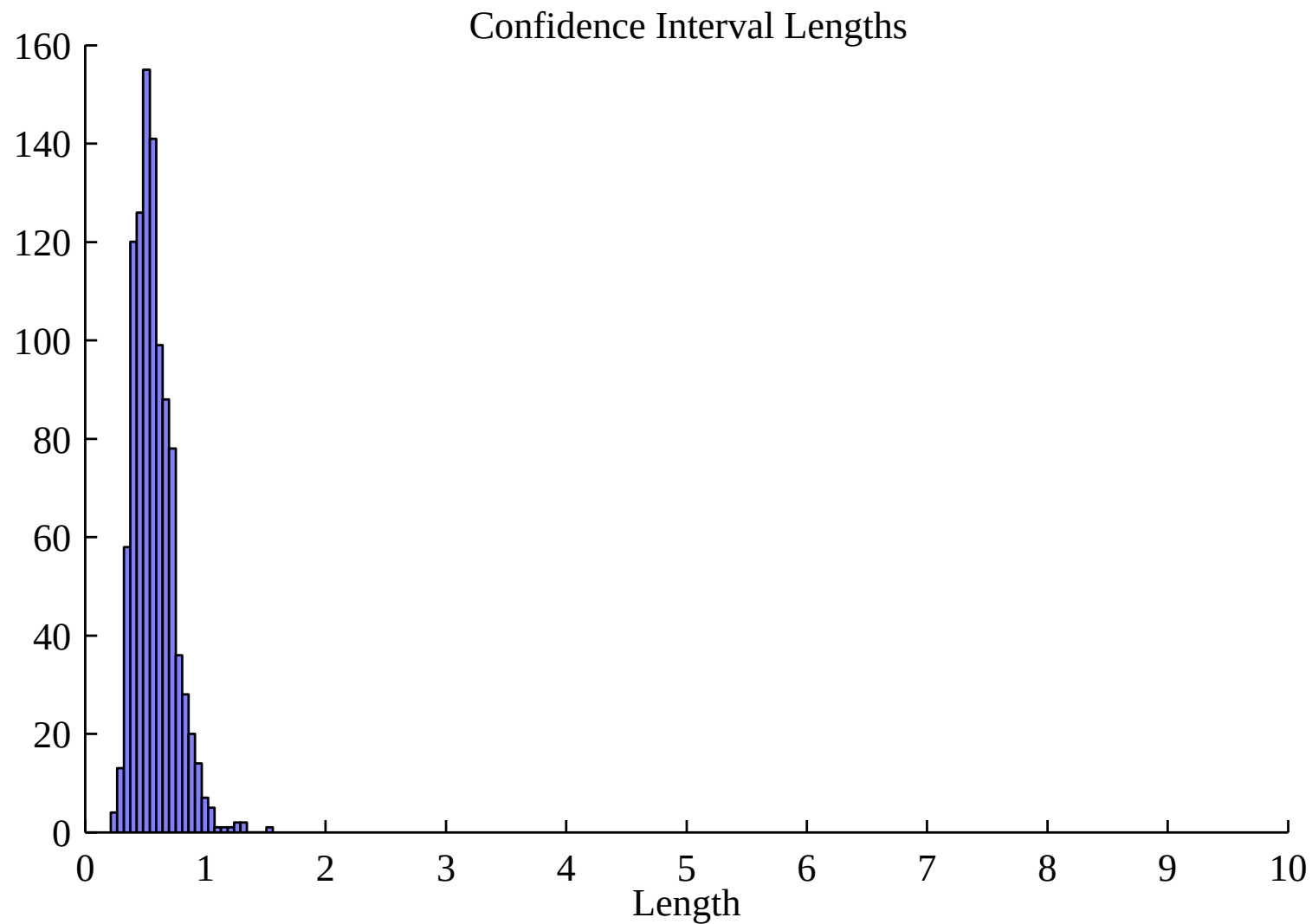
Example 3: Variance Histogram, Exponential $N = 100$



Example 3: Confidence Interval Histogram, Exponential $N = 100$



Example 3: Confidence Length Histogram, Exponential $N = 100$



Example 3: Relevant MATLAB Code

```
%function [] = TransformDomain();
close all;
clear all;

M = 1000; % No. experiments
N = 100;  % No. observations
cl = 95;  % Confidence level

%ds = 'Exponential';
%tm = 1;    % True mean
%tv = 1;    % True Variance
%X = exprnd(tm,N,M);

ds = 'Normal';
tm = 0;    % True mean
tv = 1;    % True Variance
X = randn(N,M);

sx = std(X); % Std. dev. estimate
lc = sx.^2*(N-1)/chi2inv(1-(1-cl/100)/2,N-1); % Lower confidence interval
uc = sx.^2*(N-1)/chi2inv((1-cl/100)/2,N-1); % Upper confidence interval

fprintf('Variance covered: %5.2f%%\n',100*sum(lc<tv & uc>=tv)/M,char(37));

%=====
% Plot the Variance Deviation Histogram
%=====
```

```

figure
FigureSet(1,4.5,2.8);
[n,x] = hist(sx.^2,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[0.5 0.5 1.0]);
xlim([0 5]);
title('Estimated Variance');
xlabel('Estimated Variance');
box off;
AxisSet(8);
eval(sprintf('print -depsc Var%sVarianceHistogram%03d;',ds,N));

%=====
% Plot the Confidence intervals
%=====
figure
FigureSet(1,4.5,2.8);
[n,x] = hist(uc,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[1.0 0.5 0.5]);
hold on;
    [n,x] = hist(lc,25);
    h = bar(x,n,1.0);
    set(h,'FaceColor',[0.5 1.0 0.5]);
    hold off;
xlim([0 5*tv]);
title('Estimated Confidence Intervals');
xlabel('Confidence Interval');
box off;
AxisSet(8);
eval(sprintf('print -depsc Var%sConfidenceHistogram%03d;',ds,N));

```

```

%=====
% Plot the Variance Deviation Histogram
%=====
figure
FigureSet(1,4.5,2.8);
[n,x] = hist(uc-lc,25);
h = bar(x,n,1.0);
set(h,'FaceColor',[0.5 0.5 1.0]);
xlim([0 5]);
title('Confidence Interval Lengths');
xlabel('Length');
box off;
AxisSet(8);
eval(sprintf('print -depsc Var%sLengthHistogram%03d;',ds,N));

```

Summary

- Estimators are random variables with a distribution called the *sampling distribution*
- Bias, variance, and mean square error are useful measures of performance because they only require knowledge of second order statistics of the sampling distribution
- Confidence intervals are random — not the parameter being estimated
- In many cases, it is very difficult to determine properties of the estimator (bias, variance, confidence intervals, etc.) because they often rely on unknown properties of the distribution
 - Variance of $\hat{\mu}_x$ depend on $r_x(\ell)$

Summary (Continued)

- In some cases we can obtain good approximations based on the central limit theorem or other assumptions
- It is critical to scrutinize these assumptions and determine whether they are reasonable for your application
- Monte Carlo simulations are useful for examining the sampling distribution under controlled conditions